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DEVICE FOR THE CAPTURE AND REMOVAL OF DISEASE MATERIAL FROM FLUIDS

Abstract

A device for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance coating at least a portion of the at least one inner wall of the multidirectional fluidic channel.

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Background/Summary

CROSS-REFERENCE [0001] This application is a continuation of U.S. application Ser. No. 18/749,485, filed Jun. 20, 2024, which is a continuation of U.S. application Ser. No. 16/902,745, filed Jun. 16, 2020, now U.S. Pat. No. 12,078,630, issued Sep. 3, 2024, which is a continuation of International Patent Application No. PCT/US2019/012403, filed Jan. 4, 2019, which claims the benefit of U.S. Provisional Application No. 62/614,250, filed Jan. 5, 2018, each of which is incorporated herein by reference in its entirety for all purposes.

SUMMARY

[0003] Methods and devices and systems described herein are used to capture and remove bacterial pathogens or associated toxins from fluids for the treatment of sepsis, endotoxemia, bacteremia infection, and other blood-borne diseases. Provided herein are methods, devices, and systems for removing pathogens and/or toxins from fluids by flowing fluids through a multidirectional channel that is coated with a polypeptide antibiotic and/or other materials.

[0004] Provided herein are devices, methods, and systems for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance coating at least a portion of the at least one inner wall of the multidirectional fluidic channel. In some embodiments, the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents. In some embodiments, the substance coating the channel wall comprises a fixed, covalently-bonded polypeptide antibiotic. In some embodiments, the covalently-bonded polypeptide antibiotic is a polymyxin-polymyxin B and/or polymyxin E. In some embodiments, the amount of polymyxin fixed is about 0.5 mM (milliMolar) to about 50 mM. In some embodiments, the amount of polymyxin fixed is at least about 0.5 mM. In some embodiments, the amount of polymyxin fixed is at most about 50 mM. In some embodiments, the amount of polymyxin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of polymyxin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the amount of vancomycin fixed is about 0.5 mM to about 50 mM. In some embodiments, the amount of vancomycin fixed is at least about 0.5 mM. In some embodiments, the amount of vancomycin fixed is at most about 50 mM. In some embodiments, the amount of vancomycin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1

mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of vancomycin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM. In some embodiments, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% volume/volume (v/v) aqueous solution to about 20% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is at least about 1% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is at most about 20% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% (v/v) aqueous solution to about 2% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 3% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 3% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 10% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 10% (v/v) aqueous solution to about 20% (v/v) aqueous solution, or about 15% (v/v) aqueous solution to about 20% (v/v) aqueous solution. The term about, when used in reference to the amount of the hexamethylene diamine crosslinking agent aqueous solution, means 1% (v/v) aqueous solution, 2% (v/v) aqueous solution, 3% (v/v) aqueous solution, 4% (v/v) aqueous solution, 5% (v/v) aqueous solution, 10% (v/v) aqueous solution, 15% (v/v) aqueous solution, or 20% (v/v) aqueous solution. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1 mM to about 50 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is at least about 1 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is at most about 50 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM,

about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of crosslinking agent, means 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM. In some embodiments of the devices or systems herein, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments, the thermoplastic polymer base material has at least one surface exposed functional group selected from the groups consisting of: carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride and methylmethacrylate. In some embodiments, the base material is polycarbonate.

[0005] In some embodiments of the device, the fluidic device is disposable. In some embodiments of the device, the multidirectional channel has a width of about 0.01 mm (millimeter) to about 1,000 mm. In some embodiments of the device, the multidirectional channel has a width of at least about 0.01 mm. In some embodiments of the device, the multidirectional channel has a width of at most about 1,000 mm. In some embodiments of the device, the multidirectional channel has a width of about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.01 mm to about 500 mm, about 0.01 mm to about 1,000 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm, about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.05 mm to about 500 mm, about 0.05 mm to about 1,000 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the width of the multidirectional channel, means 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, or 500 mm. In some embodiments of the device, the multidirectional channel has a height of about 0.001 mm to about 100 mm. In some embodiments of the device, the multidirectional channel has a height of at least about 0.001 mm. In some embodiments of the device, the multidirectional channel has a height of at most about 100 mm. In some embodiments of the device, the multidirectional channel has a height of about 0.001 mm to about 0.005 mm, about 0.001 mm to about 0.01 mm, about 0.001 mm to about 0.05 mm, about 0.001 mm to about 0.1 mm, about 0.001 mm to about 0.5 mm, about 0.001 mm to about 1 mm, about 0.001 mm to about 10 mm, about 0.001 mm to about 50 mm, about 0.001 mm to about 100 mm, about 0.005 mm to about 0.01 mm, about 0.005 mm to about 0.05 mm, about 0.005 mm to about 0.1 mm, about 0.005 mm to about 0.5 mm, about 0.005 mm to about 1 mm, about 0.005 mm to about 10 mm, about 0.005 mm to about 50 mm, about 0.005 mm to about 100 mm, about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm,

about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, or about 50 mm to about 100 mm. The term about, when used in reference to the height of the multidirectional channel, means 0.001 mm, 0.005 mm, 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, or 50 mm. In some embodiments of the device, the multidirectional channel has a length of about 0.1 mm to about 10,000 mm. In some embodiments of the device, the multidirectional channel has a length of at least about 0.1 mm. In some embodiments of the device, the multidirectional channel has a length of at most about 10,000 mm. In some embodiments of the device, the multidirectional channel has a length of about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, about 0.1 mm to about 5,000 mm, about 0.1 mm to about 10,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 0.5 mm to about 5,000 mm, about 0.5 mm to about 10,000 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 1 mm to about 5,000 mm, about 1 mm to about 10,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 10 mm to about 5,000 mm, about 10 mm to about 10,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 50 mm to about 5,000 mm, about 50 mm to about 10,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, about 100 mm to about 5,000 mm, about 100 mm to about 10,000 mm, about 500 mm to about 1,000 mm, about 500 mm to about 5,000 mm, about 500 mm to about 10,000 mm, about 1,000 mm to about 5,000 mm, about 1,000 mm to about 10,000 mm, or about 5,000 mm to about 10,000 mm. The term about, when used in reference to length of the multidirectional channel, means 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, 500 mm, 1,000 mm, or 5,000 mm. In some embodiments of the device, the multidirectional channel has a length of about 1.0 to about 3,000.0 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 0.1 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at least about 0.1 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at most about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 5 mm to about 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at least about 5 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at most about 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 5 mm to about 10 mm, about 5 mm to about 25 mm, about 5 mm to about 50 mm, about 5 mm to about 75 mm, about 5 mm to about 100 mm, about 10 mm to about 25 mm, about 10 mm to about 50 mm, about 10 mm to about 75 mm, about 10 mm to about 100 mm, about 25 mm to about 50 mm, about 25 mm to about 75 mm,

about 25 mm to about 100 mm, about 50 mm to about 75 mm, about 50 mm to about 100 mm, or about 75 mm to about 100 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to-center distance between the channels in the spiral about 0.01 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to-center distance between the channels in the spiral at least about 0.01 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to-center distance between the channels in the spiral at most about 1,000 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to-center distance between the channels in the spiral about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.01 mm to about 500 mm, about 0.01 mm to about 1,000 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm, about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.05 mm to about 500 mm, about 0.05 mm to about 1,000 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the distance between the channel in the spiral, means 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, or 500 mm. In some embodiments of the device, the multidirectional channel is helically shaped and fabricated around a cylindrical chamber. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of about 0.1 mm to about 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of at least about 0.1 mm. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of at most about 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of about 0.1 mm to about 5 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, 1 mm to about 5 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 5 mm to about 10 mm, about 5 mm to about 50 mm, about 5 mm to about 100 mm, about 5 mm to about 500 mm, about 5 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the radius of curvature of the helical shaped multidirectional channel, means 0.1 mm, 1 mm, 5 mm, 10 mm, 50 mm, 100 mm, 500 mm, or 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of 5.0 to 100.0 mm. In some embodiments of the device, the helical shaped multidirectional channel has a radius of curvature of about 5.0 to about 100.0 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 1 mm to about 1,000 mm, with the pitch measured as the height of one complete helix turn measured parallel to the axis of the helix. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of at least about 0.1 mm. In some embodiments of the device, the helical shaped multidirectional channel has

a pitch of at most about 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 0.1 mm to about 5 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, 1 mm to about 5 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 5 mm to about 10 mm, about 5 mm to about 50 mm, about 5 mm to about 100 mm, about 5 mm to about 500 mm, about 5 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the pitch, means 0.1 mm, 1 mm, 5 mm, 10 mm, 50 mm, 100 mm, 500 mm, or 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 1.0 to about 100.0 mm. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of about 1 mL/min (milliliters per minute) to about 1,000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of at least about 1 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of at most about 1,000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of about 1 mL/min to about 5 mL/min, about 1 mL/min to about 10 mL/min, about 1 mL/min to about 50 mL/min, about 1 mL/min to about 100 mL/min, about 1 mL/min to about 500 mL/min, about 1 mL/min to about 1,000 mL/min, about 5 mL/min to about 10 mL/min, about 5 mL/min to about 50 mL/min, about 5 mL/min to about 100 mL/min, about 5 mL/min to about 500 mL/min, about 5 mL/min to about 1,000 mL/min, about 10 mL/min to about 50 mL/min, about 10 mL/min to about 100 mL/min, about 10 mL/min to about 500 mL/min, about 10 mL/min to about 1,000 mL/min, about 50 mL/min to about 100 mL/min, about 50 mL/min to about 500 mL/min, about 50 mL/min to about 1,000 mL/min, about 100 mL/min to about 500 mL/min, about 100 mL/min to about 1,000 mL/min, or about 500 mL/min to about 1,000 mL/min. The term about, when used in reference to the flow rate, means 1 mL/min, 5 mL/min, 10 mL/min, 50 mL/min, 100 mL/min, 200 mL/min, 500 mL/min, or 1000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates with flow rates of about 50 to about 400 mL/min. In some embodiments of the device, the multidirectional channel is enclosed. In some embodiments of the device, the multidirectional channel is enclosed using bolts, adhesive, binding material, thermal expansion, resin, epoxy, an inner and/or outer sleeve, a base plate, a mandrel, cover glass, curing, extrusion welding, contact welding, high frequency welding, friction welding, laser welding, ultrasonic welding, solvent welding, or casting. In some embodiments of the device, the multidirectional channel is fabricated using at least one method selected from the group consisting of: 3-D printing, soft lithography, photolithography, injection molding, blow molding, casting, ultrasonic welding, high frequency welding, heated tool or plate welding, solvent bonding, laser welding, spin welding, infrared welding, vibration welding, adhesive bonding, machining, turning, drilling, boring, reaming, electric discharge machining, or milling. In some embodiments of the device, tubing is attached to the multidirectional channel inlet(s) and/or outlet(s) by the use of fittings, caps, or luer lock connectors.

[0006] Provided herein is a method for the capture and adsorption of materials of interest comprising bringing a sample of fluid in contact with a multidirectional, polypeptide antibiotic coated channel; adsorbing said material(s) of interest on said multidirectional channel walls; and detecting the presence or amount of said materials of interest captured within the multidirectional, polypeptide antibiotic coated channel. In some embodiments of the method, a fluid sample is brought in contact with a multidirectional, polypeptide antibiotic coated channel using a pump. In some embodiments, the pump is a peristaltic pump. In some embodiments, the pump is a syringe pump. In some embodiments of the method, a fluid sample is brought in contact with a

multidirectional, polypeptide antibiotic coated channel using a syringe. In some embodiments, the polypeptide antibiotic is polymyxin. In some embodiments, the polypeptide antibiotic is vancomycin. In some embodiments of the method, heparin, sodium citrate, or other anticoagulants, are added to the fluid sample prior to being brought in contact with a multidirectional, polypeptide antibiotic coated channel. In some embodiments, detection of the presence or amount of materials of interest captured within the multidirectional, polypeptide antibiotic coated channel is performed using a method selected from the group comprising: cell counting, MALDI-TOF MS (matrix assisted laser desorption ionization-time of flight mass spectrometry), mass spectrometry, PCR (polymerase chain reaction), biosensing, flow cytometry, and fluorescent labeling.

[0007] Provided herein is a system for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents, coating, fixed and covalently bonded to at least a portion of the at least one inner wall of the multidirectional fluidic channel. In some embodiments of the system, the substance coating the channel wall comprises a fixed, covalently-bonded polypeptide antibiotic. In some embodiments of the system, the covalently-bonded polypeptide antibiotic is polymyxin. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments of the system, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. In some embodiments of the system, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments of the system, the thermoplastic polymer base material has at least one surface exposed functional group selected from the groups consisting of: carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride and methylmethacrylate. In some embodiments of the system, the base material is polycarbonate. In some embodiments of the system, the fluidic device is disposable.

[0008] Provided herein is a method for treating a patient suspected of having bacteremia, endotoxemia or sepsis comprising contacting a sample of blood with the fluidic cartridge containing a multidirectional fluidic channel between at least one inlet and at least one outlet; adsorbing one or more materials of interest on at least one wall of the multidirectional fluidic channel; removing the one or more materials of interest from the sample of blood to produce treated blood; and returning said treated blood to said patient.

[0009] Provided herein is a method for diagnosing a patient suspected of having bacteremia, endotoxemia or sepsis comprising contacting a sample of fluid with a fluidic cartridge containing a multidirectional fluidic channel between at least one inlet and at least one outlet; adsorbing one or more materials of interest on at least one wall of the multidirectional fluidic channel; and detecting the presence or amount of materials of interest captured along the at least one wall of the multidirectional fluidic channel. In some embodiments, the method comprises identifying the captured materials of interest through a procedure selected from the group consisting of: polymerase chain reaction (PCR), fluorescence in situ hybridization (FISH), optically active microbeads, optically active nanoparticles, and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF). In some embodiments, the method comprises identifying the captured materials of interest through culturing methods. In some embodiments, the method comprises

identifying the captured materials of interest through elution of the captured materials of interest followed by a procedure selected from the group consisting of polymerase chain reaction (PCR), fluorescence in situ hybridization (FISH), optically active microbeads, optically active nanoparticles, and/or matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF), or culturing.

[0010] Provided herein is a filter plate for the capture and adsorption of blood-borne materials of interest comprising: a helical-shaped fluidic channel with at least one inlet and at least one outlet; said helical-shaped fluidic channel comprising at least one inner wall; and a substance coating at least a portion of the at least one inner wall of the helical-shaped fluidic channel. In some embodiments, the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents. In some embodiments, the substance coating the channel wall comprises a fixed, covalently-bonded polypeptide antibiotic. In some embodiments, the covalently-bonded polypeptide antibiotic is polymyxin. In some embodiments, the amount of polymyxin fixed is at least 0.5 mM. In some embodiments, the amount of polymyxin fixed is about 1.0 to about 50.0 mM. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the amount of vancomycin fixed is at least 0.5 mM. In some embodiments, the amount of vancomycin fixed is about 1.0 to about 50.0 mM. In some embodiments, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% to about 20% (v/v) aqueous solution. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1.0 to about 50.0 mM. In some embodiments, the filter plate is multidirectional. In some embodiments, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments, the thermoplastic polymer base material has at least one surface exposed functional group selected from the groups consisting of: carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride and methylmethacrylate. In some embodiments, the base material is polycarbonate. In some embodiments, the filter plate is disposable. In some embodiments, the multidirectional channel has a width of about 0.01 to about 1,000.0 mm. In some embodiments, the multidirectional channel has a width of about 1.0 to about 10.0 mm. In some embodiments, the multidirectional channel has a height of about 0.001 to about 100.0 mm. In some embodiments, the multidirectional channel has a height of about 0.1 to about 10.0 mm. In some embodiments, the multidirectional channel has a length of 0.1 to 10,000 mm. In some embodiments, the multidirectional channel has a length of about 1.0 to about 3,000.0 mm. In some embodiments, said multidirectional channel is spiral shaped. In some embodiments, the spiral shaped channel has an outer-most radius of curvature of about 1.0 to about 1,000.0 mm. In some embodiments, the spiral shaped channel has an outer-most radius of curvature of about 5.0 to about 100.0 mm. In some embodiments, the spiral shaped multidirectional channel has a distance between channels of 0.01 to 1,000 mm. In some embodiments, the spiral shaped multidirectional channel has a distance between the channels in the spiral of about 1.0 to about 10.0 mm. In some embodiments, said multidirectional channel is helically shaped and fabricated around a mandrel. In some embodiments, the helical shaped multidirectional channel has a radius of curvature of about 1.0 to about 1,000.0 mm. In some embodiments, the helical shaped multidirectional channel has a radius of curvature of about 5.0 to about 100.0 mm. In some embodiments, the helical shaped

multidirectional channel has a pitch of about 1.0 to about 1,000.0 mm. In some embodiments, the helical shaped multidirectional channel has a pitch of about 10.0 to about 100.0 mm. In some embodiments, the helical shaped multidirectional channel is formed in a stackable plate configuration. In some embodiments, the stackable plate configuration is between 1 plate and 25 plates. In some embodiments, the stackable plate configuration is between 2 plate and 23 plates. In some embodiments, the stackable plate configuration is between 3 plate and 20 plates. In some embodiments, the helical shaped multidirectional stackable plate is fabricated using: 3-D printing; soft lithography; photolithography; injection molding; blow molding; casting; machining; turning; drilling; boring; reaming; electric discharge machining (EDM); or milling. In some embodiments, the helical shaped multidirectional stackable plate configuration is assembled using bolts, adhesive, binding material, thermal expansion, resin, epoxy, an inner sleeve, an outer sleeve, a base plate, a mandrel, cover glass, curing, extrusion welding, contact welding, high frequency welding, friction welding, laser welding, ultrasonic welding, solvent welding, or casting. In some embodiments, the multidirectional channel is extruded. In some embodiments, the multidirectional channel is enclosed. In some embodiments, the multidirectional stackable plate configuration is enclosed within a cylinder. In some embodiments, the multidirectional channel is extruded around a mandrel. In some embodiments, the multidirectional channel is injection molded polycarbonate. In some embodiments, the multidirectional channel is incorporated into a housing formed around a mandrel. In some embodiments, the multidirectional channel is incorporated into an internal surface of the housing formed around a mandrel. In some embodiments, the multidirectional channel is incorporated into an external surface of the housing formed around a mandrel. In some embodiments, the multidirectional channel is incorporated into a housing formed around a mandrel and contained within a cylinder. In some embodiments, the multidirectional channel is stackable and contained within a cylinder. In some embodiments, the multidirectional channel is stackable and contained within a cylinder, wherein a first multidirectional channel is functionalized with a first polypeptide, and wherein a second multidirectional channel is functionalized with a second polypeptide. In some embodiments, the first polypeptide comprises polymyxin. In some embodiments, the second polypeptide comprises vancomycin. In some embodiments, the stackable multidirectional channels contained within a cylinder form a single device connected using a connector tube, wherein said channels capture and remove Gram-negative bacteria, Gram-positive bacteria, and endotoxins from flowing fluids within said single device. In some embodiments, the stackable multidirectional channels contained within the cylinder forming a single device can be connected to an in-line filtration system. In some embodiments, the in-line filtration system is a dialysis system.

[0011] Provided herein is a device for the capture and adsorption of blood-borne materials of interest comprising: a sheet comprising a helical-shaped fluidic channel with at least one inlet and at least one outlet; said helical-shaped fluidic channel comprising at least one inner wall; a mandrel attached to the sheet, wherein the sheet is rolled around the mandrel to create a helical shape; and a substance coating at least a portion of the at least one inner wall of the sheet comprising the helical-shaped fluidic channel. In some embodiments, the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents. In some embodiments, the substance coating the channel wall comprises a fixed, covalently-bonded polypeptide antibiotic. In some embodiments, the covalently-bonded polypeptide antibiotic is polymyxin. In some embodiments, the amount of polymyxin fixed is at least 0.5 mM. In some embodiments, the amount of polymyxin fixed is about 1.0 to about 50.0 mM. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the amount of vancomycin fixed is at least 0.5 mM. In some embodiments, the amount of

vancomycin fixed is about 1.0 to about 50.0 mM. In some embodiments, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% to about 20% (v/v) aqueous solution. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1.0 to about 50.0 mM. In some embodiments, the device is multidirectional. In some embodiments, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments, the thermoplastic polymer base material has at least one surface exposed functional group selected from the groups consisting of: carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride and methylmethacrylate. In some embodiments, the base material is polycarbonate. In some embodiments, the device is disposable. In some embodiments, the multidirectional channel has a width of about 0.01 to about 1,000.0 mm. In some embodiments, the multidirectional channel has a width of about 1.0 to about 10.0 mm. In some embodiments, the multidirectional channel has a height of about 0.001 to about 100.0 mm. In some embodiments, the multidirectional channel has a height of about 0.1 to about 10.0 mm. In some embodiments, the multidirectional channel has a length of 0.1 to 10,000 mm. In some embodiments, the multidirectional channel has a length of about 1.0 to about 3,000.0 mm. In some embodiments, said multidirectional channel is spiral shaped. In some embodiments, the spiral shaped channel has an outer-most radius of curvature of about 1.0 to about 1,000.0 mm. In some embodiments, the spiral shaped channel has an outer-most radius of curvature of about 5.0 to about 100.0 mm. In some embodiments, the spiral shaped multidirectional channel has a distance between channels of 0.01 to 1,000 mm. In some embodiments, the spiral shaped multidirectional channel has a distance between the channels in the spiral of about 1.0 to about 10.0 mm. In some embodiments, said multidirectional channel is helically shaped and fabricated around a mandrel. In some embodiments, the helical shaped multidirectional channel has a radius of curvature of about 1.0 to about 1,000.0 mm. In some embodiments, the helical shaped multidirectional channel has a radius of curvature of about 5.0 to about 100.0 mm. In some embodiments, the helical shaped multidirectional channel has a pitch of about 1.0 to about 1,000.0 mm. In some embodiments, the helical shaped multidirectional channel has a pitch of about 10.0 to about 100.0 mm. In some embodiments, the sheet is rolled around the mandrel between 1 and 100 times. In some embodiments, the sheet is rolled around the mandrel between 1 and 20 times. In some embodiments, the sheet is rolled around the mandrel between 1 and 5 times. In some embodiments, the sheet is fabricated using: 3-D printing; soft lithography; photolithography; injection molding; blow molding; casting; machining; turning; drilling; boring; reaming; electric discharge machining (EDM); or milling. In some embodiments, the device is assembled using bolts, adhesive, binding material, thermal expansion, resin, epoxy, an inner sleeve, an outer sleeve, a base plate, cover glass, curing, extrusion welding, contact welding, high frequency welding, friction welding, laser welding, ultrasonic welding, solvent welding, or casting. In some embodiments, the multidirectional channel is enclosed. In some embodiments, the device is enclosed within a cylinder. In some embodiments, a first multidirectional channel is functionalized with a first polypeptide, and wherein a second multidirectional channel is functionalized with a second polypeptide. In some embodiments, the first polypeptide comprises polymyxin. In some embodiments, the second polypeptide comprises vancomycin. In some embodiments, the device is contained within a cylinder form a single device connected using a connector tube, wherein said channels capture and remove Gram-negative bacteria, Gram-positive bacteria, and endotoxins from

flowing fluids within said single device. In some embodiments, the stackable multidirectional channels contained within the cylinder forming a single device can be connected to an in-line filtration system. In some embodiments, the in-line filtration system is a dialysis system.

INCORPORATION BY REFERENCE

[0012] All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference. This application expressly incorporates herein by reference the contents of PCT Application No. PCT/US17/41038.

Description

BRIEF DESCRIPTION OF FIGURES

[0013] The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which:

[0014] FIG. 1 is a representation of particles flowing through a channel and associated forces;

[0015] FIG. 2A illustrates adsorption of fluorescently labeled *A. baumannii* ATCC 17978 along polymyxin functionalized walls of an embodiment polycarbonate device;

[0016] FIG. 2B illustrates absence of adsorption of fluorescently labeled *A. baumannii* ATCC 17978 along PEGylated walls of an embodiment polycarbonate device;

[0017] FIG. 3A illustrates adsorption of fluorescently labeled *S. aureus* ATCC 29213 along vancomycin functionalized walls of an embodiment polycarbonate device;

[0018] FIG. 3B illustrates absence of adsorption of fluorescently labeled *S. aureus* ATCC 29213 along PEGylated walls of an embodiment polycarbonate device;

[0019] FIG. 4 illustrates chemistry of functionalization of polycarbonate surface with polymyxin E (i.e. colistin) according to the embodiments described herein;

[0020] FIG. 5 illustrates chemistry of functionalization of polycarbonate surface with vancomycin according to the embodiments described herein;

[0021] FIG. 6 illustrates an embodiment device prepared according to the description herein;

[0022] FIG. 7 illustrates an embodiment device prepared according to the description herein;

[0023] FIG. 8A illustrates a non-limiting embodiment of a spiral-shaped multidirectional channel, (channel Length=1,816 mm, Width=5.08 mm, Height=1.75 mm), fabricated using milling techniques and sealed by bolting two polycarbonate pieces together;

[0024] FIG. 8B illustrates another non-limiting embodiment of a spiral-shaped multidirectional channel, (channel Length=2,009 mm, Width=2 mm, Height=0.25 mm), fabricated using milling techniques and sealed by bolting two polycarbonate pieces together;

[0025] FIG. 9A illustrates a helical-shaped multidirectional channel fabricated around a cylindrical chamber or mandrel;

[0026] FIG. 9B illustrates a helical-shaped multidirectional channel not fabricated around a cylindrical chamber or mandrel;

[0027] FIG. 9C illustrates a more compact helical-shaped multidirectional channel not fabricated around a cylindrical chamber or mandrel;

[0028] FIG. 10A illustrates a capture of *A. baumannii* ATCC 17978 when flowed through the polymyxin coated device (i.e. colistinated device) at 80 ml min.sup.-1;

[0029] FIG. 10B illustrates the total capture capacity of the device was 2.39E7 CFUs;

[0030] FIG. 11A illustrates a capture of *K. pneumoniae* ATCC 700603 when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min.sup.-1;

[0031] FIG. **11B** illustrates the total capture capacity of the device was 1.98E7 CFUs;

[0032] FIG. **12A** illustrates the capture of colistin resistant *A. baumannii* when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min.sup.-1;

[0033] FIG. **12B** illustrates the initial capture efficiency was nearly 100%; the total capture capacity of the device was 7.89E6 CFUs;

[0034] FIG. **13A** illustrates the capture of colistin resistant *K. pneumoniae* ATCC 700603 when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min⁻¹;

[0035] FIG. **13B** illustrates the initial capture efficiency was nearly 80%. The total capture capacity of the device was 7.31E6 CFUs;

[0036] FIG. **14A** illustrates the capture of *S. aureus* ATCC 29213 when flowed through the vancomycin functionalized device at 80 ml min.sup.-1;

[0037] FIG. **14B** illustrates the total capture capacity of the device was 5.24E6 CFUs;

[0038] FIG. **15** summarizes the capture capacity of the bacteria adsorbent device using a variety of bacterial strains;

[0039] FIG. **16** illustrates that an endotoxin was spiked into endotoxin-free water (1 microgram per milliliter (ug ml.sup.-1)) and flowed through the polymyxin E coated device (i.e. colistinated device) and PEGylated double spiral fluidic devices at 80 ml min.sup.-1, the amount of endotoxin captured over time was assessed, and results were plotted as the mean±SD, n=3;

[0040] FIG. **17** illustrates a helical-shaped multidirectional channel fabricated around a cylindrical chamber or mandrel that is sealed and enclosed using an outer sleeve;

[0041] FIG. **18** illustrates the channel(s)'s inlet(s) and/or outlet(s) fitting of the channel of FIG. **17**;

[0042] FIG. **19** illustrates example embodiment wherein whole blood infected with bacteria and/or endotoxin being pumped from a patient to the inlet of the double-spiral fluidic device that is functionalized along the channel walls with polypeptide antibiotics designed to capture bacteria and endotoxin, followed by the blood then containing only healthy material being returned to the patient;

[0043] FIG. **20** illustrates example embodiment wherein whole blood infected with bacteria and/or endotoxin being pumped from a patient to the inlet of the helical fluidic devices in series;

[0044] FIG. **21A** illustrates an embodiment comprising a filter plate that is a helical-shaped channel fabricated using machining methods or injection molding, with each filter plate channel being 98 mm length at the outermost edge of curve;

[0045] FIG. **21B** illustrates an embodiment comprising multiple filter plates that are assembled, stacked, and sealed together using solvent welding and adhesive, yielding a total channel length of 1960 mm;

[0046] FIG. **21C** illustrates a cross-sectional view of embodiment stacked helical plates of FIG. **21B**;

[0047] FIG. **21D** illustrates sealed joints of embodiment stacked helical filter plates of FIG. **21B**.

[0048] FIG. **22A** illustrates an embodiment comprising stacked helical filter plates in housing and

[0049] incorporating an inlet and outlet;

[0050] FIG. **22B** illustrates an embodiment device comprising two sets of stacked helical filter plates in housing, each with different polypeptides;

[0051] FIG. **23A** illustrates an embodiment device comprising polycarbonate tubing extruded to form a helical-shaped multidirectional channel;

[0052] FIG. **23B** illustrates an embodiment device comprising polycarbonate tubing extruded to form a helical-shaped multidirectional channel that is fabricated around a mandrel;

[0053] FIG. **24A** illustrates an embodiment device made from an injection molded polycarbonate sheet attached to a mandrel with the helical-shaped channel incorporated into the sheet design;

[0054] FIG. **24B** illustrates an embodiment device made from a sheet rolled into a helical shape around a mandrel by first heating the polycarbonate flat sheet and then tightly rolling the flat sheet around the mandrel to create a helical shape wherein two of these rolled sheets are connected in

series using connector tubing (not shown);

[0055] FIG. 25A illustrates an embodiment device made from injection molded polycarbonate parts that incorporate the helical-shaped channel design into the housing, while using a solid inner mandrel;

[0056] FIG. 25B illustrates a cross-sectional view of the helical housing wrapped around the mandrel of FIG. 25A;

[0057] FIG. 25C illustrates an embodiment helical housing of FIG. 25A assembled around the mandrel using ultrasonic welding and/or solvent welding;

[0058] FIG. 26A illustrates an embodiment comprising an outer tube of cast or extruded polycarbonate and core A and core B injection molded polycarbonate with the helical channel design incorporated into the cores, wherein the faces of core A and core B can be coated with elastomer;

[0059] FIG. 26B illustrates the embodiment of FIG. 26A wherein the core halves are assembled and pressed into the outer tube with sealing through the use of an elastomer;

[0060] FIG. 27A illustrates an embodiment comprising two stacked and assembled injection molded polycarbonate parts that incorporate the helical-shaped channel design into the core pieces, while using a solid outer tube and which allow for the capture and removal of Gram-negative, Gram-positive, and endotoxins from flowing fluids within a single device;

[0061] FIG. 27B illustrates a cross-sectional view of the assembled helical-core channel device of FIG. 27A; and

[0062] FIG. 28 illustrates an embodiment device of FIG. 22A comprising stacked helical filter plates in housing and incorporating an inlet and outlet.

DETAILED DESCRIPTION

[0063] An endotoxin is a lipopolysaccharide derived from cell walls of Gram-negative bacteria. The presence of endotoxin, Gram-positive bacteria, and/or Gram-negative bacteria in blood initiates a cascade of local and systemic regulatory mechanisms that can result in sepsis. Sepsis, defined as a life-threatening organ dysfunction caused by a dysregulated host response to infection, afflicts over one million Americans annually and has an associated mortality rate ranging from 25-50%. Sepsis is the leading cause of death in the critically ill in the United States, costing the United States over \$20 billion annually. As the average life expectancy increases and the number of invasive procedures expand, the incidence of sepsis is rising. Currently, no specific sepsis treatment is available. Treatment of sepsis relies primarily on early recognition and rapid administration of antibiotics, fluid resuscitation, and vasoactive medications. Early, effective antibiotic therapy is essential and improves patient outcomes. However, sepsis-associated mortality remains unacceptably and persistently high, which highlights the urgent need for new sepsis therapies. Widespread use of antibiotics has significantly increased the number of drug resistant bacterial strains, suggesting that an extracorporeal device-based therapy may be more effective overall.

[0064] Much research has been done to evaluate experimental adjunct treatments for sepsis, such as extracorporeal cytokine filtration, recombinant human activated protein C, corticosteroids, human recombinant lactoferrin, and immunomodulation. Although immunomodulation has been widely anticipated, the heterogeneity of the patient population and the complexity of sepsis pathogenesis have limited advancement of these experimental approaches. Blockage of single mediators, such as interleukin-1 (IL-1) or tumor necrosis factor-alpha (TNF- α), has shown little promise in improving sepsis survival. Cytokine removal has displayed encouraging results in animal studies; however, results are believed to be a consequence of modulating other downstream mechanisms rather than the direct impact of cytokine removal. Mechanical bacterial removal using magnetic nanoparticles has been reported to improve survival of septic rodent models. Surface modification of nanoparticles using bacterial targeting ligands can lead to the efficient and reproducible capture of several important pathogenic bacteria. However, these approaches suffer from potential limitations in scale-up for treatment of large living systems and uncertain regulatory hurdles in regard to blood

contact with nanoparticles.

[0065] Provided herein are devices, systems and methods that are a clinically translatable, alternative therapy for sepsis that can remove both the pathogenic bacteria and modulate the resulting molecular effectors to help mitigate the systemic inflammatory response characteristic of sepsis and inhibit sepsis progression. Methods and devices and systems described herein are used to capture and remove bacterial pathogens or associated toxins from fluids for the treatment of sepsis, endotoxemia, bacteremia infection, and other blood-borne diseases. Provided herein are methods, devices, and systems for removing pathogens and/or toxins from fluids by flowing fluids through a multidirectional channel that is coated with a polypeptide antibiotic and/or other materials.

[0066] Extracorporeal blood cleansing technologies for sepsis have been tested by others with little success. These technologies, in the past, have used dialysis membranes, fibers, or porous beads to capture inflammatory molecules based on size or molecular weight. However, bacteria and many other sepsis mediators are too large to be removed from the blood using these technologies. Furthermore, current extracorporeal blood cleansing technologies are prone to clogging and indiscriminate capture of healthy blood cell components. Polymyxin E, also known as colistin, is a cationic polypeptide antibiotic used for the treatment of Gram-negative infection and for detoxifying endotoxin. The positive charge of polymyxins, or colistin, allows for binding to the negatively charged outer membrane of Gram-negative pathogens and endotoxin. Although it is a powerful antibiotic, its use is limited due to its nephro- and neuro-toxicity. Therefore, its intravenous use is limited. It is known that polymyxin immobilized on beads can absorb endotoxin. However, the use of beads, nanoparticles, porous materials, and fibers has proven inefficient in the capture of endotoxin from fluids such as blood. Vancomycin is a polypeptide antibiotic that can be used for the treatment of Gram-positive infections. Vancomycin interacts with the cell wall of Gram-positive pathogens and inhibits bacterial cell wall assembly. This leads to activation of bacterial autolysins that destroy the cell wall by lysis. However, vancomycin is nephrotoxic when administered systemically. Also, none of these extracorporeal blood cleansing technologies for sepsis technologies are capable of efficient capture and adsorption of blood-borne Gram-positive and Gram-negative pathogens, both of which are root causes of sepsis.

[0067] Provided herein are devices, methods, and systems for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance coating at least a portion of the at least one inner wall of the multidirectional fluidic channel. In some embodiments, the substance coating the channel wall is fixed to the channel wall. In some embodiments, the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents. In some embodiments, the substance coating the channel wall comprises a covalently-bonded polypeptide antibiotic. In some embodiments, the covalently-bonded polypeptide antibiotic is a polymyxin-polymyxin B and/or polymyxin E. In some embodiments, the covalently-bonded polypeptide antibiotic is a polymyxin-polymyxin B and/or polymyxin E. In some embodiments, the amount of polymyxin fixed is about 0.5 mM to about 50 mM. In some embodiments, the amount of polymyxin fixed is at least about 0.5 mM. In some embodiments, the amount of polymyxin fixed is at most about 50 mM. In some embodiments, the amount of polymyxin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5

mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of polymyxin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM. In some embodiments, the fixed, covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the substance coating the channel wall comprises a covalently-bonded polypeptide antibiotic. Therefore, in some embodiments, the fixed, covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the covalently-bonded polypeptide antibiotic is fixed to the channel wall. In some embodiments, the substance coating the channel wall is fixed to the channel wall. The amount of vancomycin fixed is about 0.5 mM to about 50 mM. In some embodiments, the amount of vancomycin fixed is at least about 0.5 mM. In some embodiments, the amount of vancomycin fixed is at most about 50 mM. In some embodiments, the amount of vancomycin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of vancomycin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM.

[0068] As used herein, and unless otherwise specified, the phrase “material of interest”, “materials of interest” or “disease material” means captured disease material being one or more disease materials comprising cancer cells, circulating tumor cells, peptides, beta amyloid, proteins, enzymes, toxins, diseased cells, infectious microorganisms, cells, parasites, fungi, viruses, microorganisms, bacteria, bacterial toxin, quorum sensing proteins or receptors, lipopolysaccharide, cytokines, IL-1 β , IL-4, IL-6, IL-8, IL-10, IL-11, IL-13, IL-15, IL-16, tumor necrosis factors, procalcitonin, pathogen-associated molecular patterns, C reactive protein, or a small or protein bound biological molecule relevant to liver failure.

[0069] As used herein, and unless otherwise specified, the term “about” or “approximately” means an acceptable error for a particular value as determined by one of ordinary skill in the art, which depends in part on how the value is measured or determined. In certain embodiments, the term “about” or “approximately” means within 1, 2, 3, or 4 standard deviations. In certain embodiments, the term “about” or “approximately” means within 30%, 25%, 20%, 15%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1%, 0.5%, 0.1%, or 0.05% of a given value or range. In certain embodiments, the term “about” or “approximately” means within 40.0 mm, 30.0 mm, 20.0 mm, 10.0 mm 5.0 mm 1.0 mm, 0.9 mm, 0.8 mm, 0.7 mm, 0.6 mm, 0.5 mm, 0.4 mm, 0.3 mm, 0.2 mm or 0.1 mm of a given value or range.

[0070] As used herein, the terms “comprises”, “comprising”, or any other variation thereof, are intended to cover a nonexclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

[0071] As used herein, the terms “user”, “subject” or “patient” are used interchangeably. As used herein, the terms “subject” and “subjects” refers to an animal (e.g., birds, reptiles, and mammals), a mammal including a primate (e.g., a monkey, chimpanzee, and a human) and a non-primate (e.g., a

camel, donkey, zebra, cow, pig, horse, cat, dog, rat, and mouse). In certain embodiments, the mammal is 0 to 6 months old, 6 to 12 months old, 1 to 5 years old, 5 to 10 years old, 10 to 15 years old, 15 to 20 years old, 20 to 25 years old, 25 to 30 years old, 30 to 35 years old, 35 to 40 years old, 40 to 45 years old, 45 to 50 years old, 50 to 55 years old, 55 to 60 years old, 60 to 65 years old, 65 to 70 years old, 70 to 75 years old, 75 to 80 years old, 80 to 85 years old, 85 to 90 years old, 90 to 95 years old or 95 to 100. In some embodiments, the subject or patient is a pig. In certain embodiments, the pig is 0 to 6 months old, 6 to 12 months old, 1 to 5 years old, 5 to 10 years old or 10 to 15 years old. The natural lifespan of a pig is 10-15 years.

[0072] As used herein, and unless otherwise specified, the term “transverse force”, (also called the Euler force) is the tangential force that is felt in reaction to any angular acceleration. Also known as azimuthal acceleration or transverse acceleration. It is an acceleration that appears when a non-uniformly rotating reference frame is used for analysis of motion and there is variation in the angular velocity of the reference frame's axes. This definition is typically restricted to a reference frame that rotates about a fixed axis. The Euler force is related to the Euler acceleration by “ $F=ma$ ”, where “a” is the Euler acceleration and m is the mass of the body. Said another way, the Euler force will be felt by a person riding a merry-go-round. As the ride starts, the Euler force will be the apparent force pushing the person to the back of the horse, and as the ride comes to a stop, it will be the apparent force pushing the person towards the front of the horse. A person on a horse close to the perimeter of the merry-go-round will perceive a greater apparent force than a person on a horse closer to the axis of rotation. The Euler force is perpendicular to the centrifugal force and is in the plane of rotation.

[0073] As used herein, the terms “therapies” and “therapy” can refer to any protocol(s), method(s), compositions, formulations, and/or agent(s) that can be used in the prevention, treatment, management, or amelioration of a condition (e.g., a virus infection or a condition or symptom associated therewith, an infection other than a virus infection or a condition or symptom associated therewith, an IFN treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition). In certain embodiments, the terms “therapies” and “therapy” refer to biological therapy, supportive therapy, and/or other therapies useful in treatment, management, prevention, or amelioration of a virus infection or a condition or symptom associated therewith, an infection other than a virus infection or a condition or symptom associated therewith, an IFN treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition, known to one of skill in the art.

[0074] As used herein, the terms “manage,” “managing,” and “management” refer to the beneficial effects that a subject derives from a therapy (e.g., a prophylactic or therapeutic agent), which does not result in a cure of the disease. In certain embodiments, a subject is administered one or more therapies (e.g., one or more prophylactic or therapeutic agents) to “manage” a disease so as to prevent the progression or worsening of the disease.

[0075] As used herein, the terms “prevent”, “preventing” and “prevention” refer to the inhibition of the development or onset of a condition (e.g.: a virus infection or a condition associated therewith, an infection other than a virus infection or a condition associated therewith, an IFN-treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition), or the prevention of the recurrence, onset, or development of one or more symptoms of a condition (e.g., virus infection or a condition associated therewith, an infection other than a virus infection or a condition associated therewith, or an IFN-treatable disease), in a subject resulting from the administration of a therapy (e.g., a prophylactic or therapeutic agent), or the administration of a combination of therapies (e.g., a combination of prophylactic or therapeutic agents).

[0076] As used herein, the terms “treat,” “treatment,” and “treating” refer to the eradication or control of virus replication or the replication of a pathogen (e.g., a virus, a bacteria) other than

virus, the reduction in the titer of virus or titer other than virus, the reduction in the numbers of a pathogen, the reduction or amelioration of the progression, severity, and/or duration of a condition (e.g., a virus infection or a condition associated therewith, an infection other than a virus infection or a condition or symptom associated therewith, an IFN treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition), or the amelioration of one or more symptoms resulting from the administration of one or more therapies (including, but not limited to, the administration of one or more prophylactic or therapeutic agents).

[0077] As used herein, the terms “therapeutic agent” and “therapeutic agents” refer to any agent(s) which can be used in the prevention, treatment, management, or amelioration of a condition or a symptom thereof (e.g., an infection or a condition or symptoms associated therewith, an infection other than a virus infection or a condition or symptom associated therewith, an IFN treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition). In some embodiments, a therapeutic agent is an agent which is known to be useful for, or has been or is currently being used for the prevention, treatment, management, or amelioration of a virus infection or a condition or symptoms associated therewith, an infection other than a virus infection or a condition or symptom associated therewith, an IFN treatable disease or a condition in which the attenuated viruses can be used as a vector to induce an immune response to a particular antigen associated with the condition.

[0078] As used herein, and unless otherwise specified, the term “helix” means an object having a three-dimensional shape like that of a wire wound uniformly in a single layer around a cylinder or cone, as in a corkscrew or spiral staircase. Alternative synonymous terms may comprise spiral corkscrew, curl, curlicue, twist, gyre, whorl, convolution, etc. Geometrically a helix, as used herein can also mean a curve on a conical or cylindrical surface that would become a straight line if the surface were unrolled into a plane.

[0079] As used herein, and unless otherwise specified, the term “cylindrical” means having straight parallel sides and a circular or oval cross-section; in the shape or form of a cylinder; i.e.: “a cylindrical plastic container”.

[0080] As used herein, and unless otherwise specified, the term “multidirectional” means involving, or operating in several directions, such as moving around in a curved, non-linear or radial direction, such as around a column, a cylinder, a cone, a barrel shape, a parabolic, an ellipse, a hyperbola, etc., as non-limiting examples.

[0081] Provided herein are devices, methods, and systems for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance coating at least a portion of the at least one inner wall of the multidirectional fluidic channel.

[0082] The presently described embodiments relate to a membrane-free pathogen and endotoxin adsorption device technology that is capable of continuous flow and high throughput. The working principle relies on two components: 1) polypeptide antibiotics bonded to the device channel walls and 2) fluidic flow in curved or multidirectional channel structures, such as a spiral or helix, eliminating the need for membrane-based filters or external forces. FIG. 1 is a representation of particles flowing through a channel and associated forces. Transverse force in a multidirectional channel, such as a spiral or helix, concentrates particulates according to the designed channel aspect ratio (FIG. 1). Transverse Dean flows exert a Dean drag force ($F_{sub.D}$) on particles flowing in a spiral channel. The competition between wall lift force, shear lift force, and Dean drag force results in differential migration of particles to unique equilibrium positions dependent of particle size (Johnston, I. D. et al. Dean flow focusing and separation of small microspheres within a narrow size range. *Microfluid. Nanofluidics* 17, 509-518 (2014)). Concentrating particulates, such as bacteria and endotoxin, near the multidirectional channel sidewalls functionalized with

polypeptide antibiotics promotes capture, adsorption, and removal of these particulates from fluid through interaction with the polypeptide antibiotics. The simplicity of this design (i.e. no moving parts, no membrane-based filter, etc.) allows for incorporation of this device into other downstream processes, such as hemodialysis. This device can also serve as stand-alone blood filtration device for the diagnosis and treatment of bacteremia, endotoxemia, sepsis, and other blood-borne diseases. [0083] The described embodiments use a curved channel of a spiral or helical device to introduce centrifugal force upon particulates (i.e. bacteria and endotoxin) flowing in a fluid (i.e. blood, saline, urine) to facilitate improved separation, capture, and removal of such particulates from the fluid. As these particulates flow through the channel, the particulates are forced toward the channel sidewalls, in a position offset from the center of the channel. The combination of centrifugal and fluidic forces in a helical- or spiral-shaped multidirectional channel allows for concentration of particulates near the polypeptide antibiotic functionalized sidewalls for capture and adsorption of bacteria and endotoxin.

[0084] The polypeptide antibiotics described herein, including polymyxin and vancomycin, are antibiotic substances having a strong affinity for the outer membrane of various strains of Gram-positive and Gram-negative bacteria, as well as an affinity for endotoxins. This allows for capture and adsorption of these biologics.

[0085] In some embodiments of the devices or systems herein, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments, the base material is polycarbonate. In some embodiments of the devices or systems herein, the multidirectional channel is made of a base material that is typically a thermoplastic polymer, such as polycarbonate. The thermoplastic polymer, in some embodiments, has surface exposed functional groups capable of fixing polymyxin, vancomycin, and other relative molecules. Examples of the exposed functional groups characteristic of the base material include carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride, methylmethacrylate, etc.

[0086] In some embodiments, the substance coating the channel wall is fixed to the channel wall. In some embodiments, the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents.

[0087] Among the materials listed, polycarbonate derivatives are suitable for the carrier due to the exposed carbonyl groups and biocompatibility. The exposed carbonyl groups on the polycarbonate surface undergo nucleophilic addition by the amine groups of the polypeptide antibiotic, such as colistin or vancomycin. This reaction results in scission of the polymer chain and the formation of a terminal carbamate. Addition of the polypeptide antibiotic containing one or more amine groups results in a polycarbonate surface decorated with the polypeptide antibiotic, which is available for subsequent capture of bacteria and/or endotoxin (FIG. 4 and FIG. 5).

[0088] FIG. 4 illustrates chemistry of functionalization of polycarbonate surface with polymyxin E (i.e. colistin) according to the embodiments described herein. FIG. 4 illustrates chemistry of functionalization of polycarbonate surface with polymyxin E (i.e. colistin) according to the embodiments described herein. In some embodiments, the substance coating the channel wall comprises a covalently-bonded polypeptide antibiotic. In some embodiments, the covalently-bonded polypeptide antibiotic is a polymyxin-polymyxin B and/or polymyxin E. In some embodiments, the covalently-bonded polypeptide antibiotic is a polymyxin-polymyxin B and/or polymyxin E. In some embodiments, the amount of polymyxin fixed is about 0.5 mM to about 50 mM. In some embodiments, the amount of polymyxin fixed is at least about 0.5 mM. In some embodiments, the amount of polymyxin fixed is at most about 50 mM. In some embodiments, the

amount of polymyxin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of polymyxin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM.

[0089] FIG. 5 illustrates chemistry of functionalization of polycarbonate surface with vancomycin according to the embodiments described herein. FIG. 5 illustrates chemistry of functionalization of polycarbonate surface with vancomycin according to the embodiments described herein. In some embodiments, the fixed, covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the substance coating the channel wall comprises a covalently-bonded polypeptide antibiotic. Therefore, in some embodiments, the fixed, covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments, the covalently-bonded polypeptide antibiotic is fixed to the channel wall. In some embodiments, the substance coating the channel wall is fixed to the channel wall. The amount of vancomycin fixed is about 0.5 mM to about 50 mM. In some embodiments, the amount of vancomycin fixed is at least about 0.5 mM. In some embodiments, the amount of vancomycin fixed is at most about 50 mM. In some embodiments, the amount of vancomycin fixed is about 0.5 mM to about 1 mM, about 0.5 mM to about 5 mM, about 0.5 mM to about 10 mM, about 0.5 mM to about 20 mM, about 0.5 mM to about 30 mM, about 0.5 mM to about 40 mM, about 0.5 mM to about 50 mM, about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of vancomycin, means 0.5 mM, 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM.

[0090] Alternatively, other functional groups conjugated to the base material surface could be attached using similar chemistry. For example, a glycine or similar structure could be used to generate a carboxylated surface. Also, hexamethylene diamine can be functionalized to the surface of the polycarbonate by nucleophilic addition by the diamine, resulting in the formation of a terminal hexylaminocarbamate. NHS esters (N-hydroxysuccinimide esters), such as NHS-Polyethylene glycol-NHS (NHS-PEG-NHS) can then be used as a cross linker and react with the amine groups exposed on the hexylaminocarbamate surface using carbodiimide chemistry. This reaction results in a surface exposed NHS group, which can then be used to react with the exposed amine groups of polypeptide antibiotics, such as vancomycin or polymyxin, forming a polypeptide functionalized surface (FIG. 6 and FIG. 7). This functionalization methodology allows for extension of the polypeptide antibiotic further from the polycarbonate surface, thereby potentially improving interaction, capture, and adsorption of target biologics, such as bacteria and endotoxin. A nitrogen atom bonded to the base material is also effective for capture and removal of Gram-negative pathogens and endotoxin.

[0091] The amount of polypeptide antibiotic substance to be fixed to the base material is, in some embodiments, more than 0.5 mM of substance and in some embodiments is from 1.0 mM to 50.0

mM substance. If the amount is less than 0.5 mM of substance, detoxifying may become ineffective, in some embodiments. In some embodiments, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. The amount of crosslinking agent, such as NHS-PEH-NHS, to be fixed to the base material is, in some embodiments, more than 0.1 mM of substance and in some embodiments, 1.0-50.0 mM substance.

[0092] FIG. 6 illustrates chemistry of functionalization of polycarbonate surface with a crosslinking agent and polymyxin E (i.e. colistin) according to the embodiments described herein. FIG. 7 illustrates chemistry of functionalization of polycarbonate surface with a crosslinking agent and vancomycin according to the embodiments described herein. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1 mM to about 50 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is at least about 1 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is at most about 50 mM. In some embodiments, the crosslinking agent to be fixed to the base material is polyethylene glycol or a derivative substance in which the amount fixed is about 1 mM to about 5 mM, about 1 mM to about 10 mM, about 1 mM to about 20 mM, about 1 mM to about 30 mM, about 1 mM to about 40 mM, about 1 mM to about 50 mM, about 5 mM to about 10 mM, about 5 mM to about 20 mM, about 5 mM to about 30 mM, about 5 mM to about 40 mM, about 5 mM to about 50 mM, about 10 mM to about 20 mM, about 10 mM to about 30 mM, about 10 mM to about 40 mM, about 10 mM to about 50 mM, about 20 mM to about 30 mM, about 20 mM to about 40 mM, about 20 mM to about 50 mM, about 30 mM to about 40 mM, about 30 mM to about 50 mM, or about 40 mM to about 50 mM. The term about, when used in reference to the amount of crosslinking agent, means 1 mM, 5 mM, 10 mM, 20 mM, 30 mM, 40 mM, or 50 mM. The amount of crosslinking agent, such as hexamethylene diamine, to be fixed to the base material is, in some embodiments, more than 0.1% aqueous hexamethylene diamine solution and in some embodiments is from 1% to 20% aqueous hexamethylene diamine solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% (v/v) aqueous solution to about 20% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is at least about 1% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is at most about 20% (v/v) aqueous solution. In some embodiments, the hexamethylene diamine crosslinking agent to be fixed to the base material is about 1% (v/v) aqueous solution to about 2% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 3% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 1% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 3% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 2% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 4% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 3% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 5% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 4% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 4%

(v/v) aqueous solution to about 20% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 10% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 5% (v/v) aqueous solution to about 20% (v/v) aqueous solution, about 10% (v/v) aqueous solution to about 15% (v/v) aqueous solution, about 10% (v/v) aqueous solution to about 20% (v/v) aqueous solution, or about 15% (v/v) aqueous solution to about 20% (v/v) aqueous solution. The term about, when used in reference to the amount of the hexamethylene diamine crosslinking agent aqueous solution, means 1% (v/v) aqueous solution, 2% (v/v) aqueous solution, 3% (v/v) aqueous solution, 4% (v/v) aqueous solution, 5% (v/v) aqueous solution, 10% (v/v) aqueous solution, 15% (v/v) aqueous solution, or 20% (v/v) aqueous solution.

[0093] Embodiments of a multidirectional fluidic channel as described herein may comprise various geometries including without limitation those illustrated in FIGS. **8A, 8B, 9A, 9B, 9C, 17, 18, 21A, 21B, 21C, 21D, 22A, 22B, 23A, 23B, 24A, 24B, 25A, 25B, 25C, 26A, 26B, 27A, 27B, and 28**. The multidirectional channel, in some embodiments, has an internal channel surface area of 0.001 square meters (m.sup.2) to 100 m.sup.2. The multidirectional channel, in some embodiments, has a surface area of from 0.001 m.sup.2 to 10 m.sup.2, as well as exposed functional groups being capable of fixing polymyxin and vancomycin. A surface area too large (beyond the maximum of 100 m.sup.2 or 10 m.sup.2, for example depending on the embodiment) increases the pressure drop of the fluid (e.g. blood) passed through the device too much, and a surface area that is too small (below 0.001 m.sup.2) makes the detoxifying capacity of the device insufficient in removing the bacteria and/or pathogens, at least. The pressure drop is in some embodiments, less than 10 pounds per square inch (psi) [69 kilopascals (kPa)], and in some embodiments less than 3 psi [21 kPa].

[0094] The internal channel surface area may comprise the area of the entire surface coated by the substance. The multidirectional channel, in some embodiments, has an internal channel surface area of 0.001 m.sup.2 to 80 m.sup.2, 0.001 m.sup.2 to 70 m.sup.2, 0.001 m.sup.2 to 60 m.sup.2, 0.001 m.sup.2 to 50 m.sup.2, 0.001 m.sup.2 to 40 m.sup.2, 0.001 m.sup.2 to 30 m.sup.2, 0.001 m.sup.2 to 25 m.sup.2, 0.001 m.sup.2 to 20 m.sup.2, 0.001 m.sup.2 to 19 m.sup.2, 0.001 m.sup.2 to 18 m.sup.2, 0.001 m.sup.2 to 17 m.sup.2, 0.001 m.sup.2 to 16 m.sup.2, 0.001 m.sup.2 to 15 m.sup.2, 0.001 m.sup.2 to 14 m.sup.2, 0.001 m.sup.2 to 13 m.sup.2, 0.001 m.sup.2 to 12 m.sup.2, 0.001 m.sup.2 to 11 m.sup.2, 0.01 m.sup.2 to 20 m.sup.2, 0.01 m.sup.2 to 19 m.sup.2, 0.01 m.sup.2 to 18 m.sup.2, 0.01 m.sup.2 to 17 m.sup.2, 0.01 m.sup.2 to 16 m.sup.2, 0.01 m.sup.2 to 15 m.sup.2, 0.01 m.sup.2 to 14 m.sup.2, 0.01 m.sup.2 to 13 m.sup.2, 0.01 m.sup.2 to 12 m.sup.2, 0.01 m.sup.2 to 11 m.sup.2, 0.01 m.sup.2 to 10 m.sup.2, 0.01 m.sup.2 to 100 m.sup.2, 0.01 m.sup.2 to 80 m.sup.2, 0.01 m.sup.2 to 70 m.sup.2, 0.01 m.sup.2 to 60 m.sup.2, 0.01 m.sup.2 to 50 m.sup.2, 0.01 m.sup.2 to 40 m.sup.2, 0.01 m.sup.2 to 30 m.sup.2, 0.01 m.sup.2 to 25 m.sup.2, 0.1 m.sup.2 to 20 m.sup.2, 0.1 m.sup.2 to 19 m.sup.2, 0.1 m.sup.2 to 18 m.sup.2, 0.1 m.sup.2 to 17 m.sup.2, 0.1 m.sup.2 to 16 m.sup.2, 0.1 m.sup.2 to 15 m.sup.2, 0.1 m.sup.2 to 14 m.sup.2, 0.1 m.sup.2 to 13 m.sup.2, 0.1 m.sup.2 to 12 m.sup.2, 0.1 m.sup.2 to 11 m.sup.2, 0.1 m.sup.2 to 10 m.sup.2, 0.1 m.sup.2 to 100 m.sup.2, 0.1 m.sup.2 to 80 m.sup.2, 0.1 m.sup.2 to 70 m.sup.2, 0.1 m.sup.2 to 60 m.sup.2, 0.1 m.sup.2 to 50 m.sup.2, 0.1 m.sup.2 to 40 m.sup.2, 0.1 m.sup.2 to 30 m.sup.2, 0.1 m.sup.2 to 25 m.sup.2, 1 m.sup.2 to 20 m.sup.2, 1 m.sup.2 to 19 m.sup.2, 1 m.sup.2 to 18 m.sup.2, 1 m.sup.2 to 17 m.sup.2, 1 m.sup.2 to 16 m.sup.2, 1 m.sup.2 to 15 m.sup.2, 1 m.sup.2 to 14 m.sup.2, 1 m.sup.2 to 13 m.sup.2, 1 m.sup.2 to 12 m.sup.2, 1 m.sup.2 to 11 m.sup.2, 1 m.sup.2 to 10 m.sup.2, 1 m.sup.2 to 100 m.sup.2, 1 m.sup.2 to 80 m.sup.2, 1 m.sup.2 to 70 m.sup.2, 1 m.sup.2 to 60 m.sup.2, 1 m.sup.2 to 50 m.sup.2, 1 m.sup.2 to 40 m.sup.2, 1 m.sup.2 to 30 m.sup.2, 1 m.sup.2 to 25 m.sup.2, about 0.001 m.sup.2 to about 100 m.sup.2, or about 0.001 m.sup.2 to about 10 m.sup.2.

[0095] The radius of curvature may be the radius of the circular arc which best approximates the curve of the coated surface of channel at a point on the outer-surface of the channel (in the direction of the Dean's wall-induced lift force). The helical-shaped multidirectional channel in some embodiments has a radius of curvature of 0.1 mm to 1,000 mm, in some embodiments has a radius

of curvature of 10 mm to 500 mm, and in some embodiments has a radius of curvature of 5 mm to 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 0.1 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at least about 0.1 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature (in the direction of the Dean's wall-induced lift force) of at most about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 5 mm to about 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at least about 5 mm. The spiral-shaped multidirectional channel in some embodiments, has an outer-most radius of curvature of 0.1 mm to 1,000 mm, and in some embodiments, the spiral-shaped multidirectional channel has an outer-most radius of curvature of 1 mm to 500 mm, and in some embodiments, the spiral-shaped multidirectional channel has an outer-most radius of curvature of 5 mm to 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of at most about 100 mm. In some embodiments of the device, the spiral shaped channel has an outer-most radius of curvature of about 5 mm to about 10 mm, about 5 mm to about 25 mm, about 5 mm to about 50 mm, about 5 mm to about 75 mm, about 5 mm to about 100 mm, about 10 mm to about 25 mm, about 10 mm to about 50 mm, about 10 mm to about 75 mm, about 10 mm to about 100 mm, about 25 mm to about 50 mm, about 25 mm to about 75 mm, about 25 mm to about 100 mm, about 50 mm to about 75 mm, about 50 mm to about 100 mm, or about 75 mm to about 100 mm.

[0096] The spiral-shaped multidirectional channel in some embodiments has a distance between channels (defined from center of channel lumen to center of channel lumen) of 0.01 mm to 1,000 mm, and in some embodiments the spiral-shaped multidirectional channel has a distance between channels of 0.1 mm to 100 mm, and in some embodiments the spiral-shaped multidirectional channel has a distance between channels of 1 mm to 10 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to-center distance between the channels in the spiral about 0.01 mm to about 1,000 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to center distance between the channels in the spiral at least about 0.01 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to center distance between the channels in the spiral at most about 1,000 mm. In some embodiments of the device, the spiral shaped multidirectional channel has a center-to center distance between the channels in the spiral about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.01 mm to about 500 mm, about 0.01 mm to about 1,000 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm, about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.05 mm to about 500 mm, about 0.05 mm to about 1,000 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 1 mm to about 10 mm, about 1 mm to

about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the distance between the channel in the spiral, means 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, or 500 mm. In some embodiments of the device, the multidirectional channel is helically shaped and fabricated around a cylindrical chamber.

[0097] The pitch of the helical channel may be defined from center of channel lumen to center of channel lumen in the cylindrical axis (e.g. non-radially symmetric axis) of helical devices. The device may be cylindrical helical or conical helical. The pitch of the helical channel is in some embodiments 0.1 mm to 1,000 mm and in some embodiments has a pitch of 1 mm to 500 mm, and in some embodiments has a pitch of 1 mm to 100 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 1 mm to about 1,000 mm, with the pitch measured as the height of one complete helix turn measured parallel to the axis of the helix. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of at least about 0.1 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of at most about 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 0.1 mm to about 5 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, 1 mm to about 5 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 5 mm to about 10 mm, about 5 mm to about 50 mm, about 5 mm to about 100 mm, about 5 mm to about 500 mm, about 5 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the pitch, means 0.1 mm, 1 mm, 5 mm, 10 mm, 50 mm, 100 mm, 500 mm, or 1,000 mm. In some embodiments of the device, the helical shaped multidirectional channel has a pitch of about 1.0 to about 100.0 mm.

[0098] In some embodiments, the cross-section of the lumen of the channel may be polygonal (e.g. square, rectangle, hexagonal, etc.) or may be ellipsoid. The channel width may be the interior surface to surface distance of the channel lumen at its widest point. The helical- and spiral-shaped multidirectional channel(s) in some embodiments have a channel width of 0.01 mm to 1,000 mm, in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel width of 0.1 mm to 100 mm, and in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel width of 1 mm to 10 mm. The channel height may be the widest, interior center to center distance within the lumen perpendicular to its widest point. The helical- and spiral-shaped multidirectional channel(s) in some embodiments have a channel height of 0.001 mm to 100 mm, and in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel height of 0.01 mm to 20 mm, and in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel height of 0.1 mm to 10 mm. The helical- and spiral-shaped multidirectional channel(s) in some embodiments have a channel length of 0.1 mm to 10,000 mm, and in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel length of 1 mm to 5,000 mm, and in some embodiments the helical- and spiral-shaped multidirectional channel(s) have a channel length of 1 mm to 3,000 mm.

[0099] In some embodiments, the cross-section of the lumen of the channel may be polygonal (e.g. square, rectangle, hexagonal, etc.) or may be ellipsoid. The channel width may be the interior surface to surface distance of the channel lumen at its widest point. In some embodiments of the device, the multidirectional channel has a width of about 0.01 mm to about 1,000 mm. In some

embodiments of the device, the multidirectional channel has a width of at least about 0.01 mm. In some embodiments of the device, the multidirectional channel has a width of at most about 1,000 mm. In some embodiments of the device, the multidirectional channel has a width of about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.01 mm to about 500 mm, about 0.01 mm to about 1,000 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm, about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.05 mm to about 500 mm, about 0.05 mm to about 1,000 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to about 500 mm, about 0.1 mm to about 1,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, or about 500 mm to about 1,000 mm. The term about, when used in reference to the width of the multidirectional channel, means 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, or 500 mm.

[0100] The channel height may be the widest, interior center-to-center distance within the lumen perpendicular to its widest point. In some embodiments of the device, the multidirectional channel has a height of about 0.001 mm to about 100 mm. In some embodiments of the device, the multidirectional channel has a height of at least about 0.001 mm. In some embodiments of the device, the multidirectional channel has a height of at most about 100 mm. In some embodiments of the device, the multidirectional channel has a height of about 0.001 mm to about 0.005 mm, about 0.001 mm to about 0.01 mm, about 0.001 mm to about 0.05 mm, about 0.001 mm to about 0.1 mm, about 0.001 mm to about 0.5 mm, about 0.001 mm to about 1 mm, about 0.001 mm to about 10 mm, about 0.001 mm to about 50 mm, about 0.001 mm to about 100 mm, about 0.005 mm to about 0.01 mm, about 0.005 mm to about 0.05 mm, about 0.005 mm to about 0.1 mm, about 0.005 mm to about 0.5 mm, about 0.005 mm to about 1 mm, about 0.005 mm to about 10 mm, about 0.005 mm to about 50 mm, about 0.005 mm to about 100 mm, about 0.01 mm to about 0.05 mm, about 0.01 mm to about 0.1 mm, about 0.01 mm to about 0.5 mm, about 0.01 mm to about 1 mm, about 0.01 mm to about 10 mm, about 0.01 mm to about 50 mm, about 0.01 mm to about 100 mm, about 0.05 mm to about 0.1 mm, about 0.05 mm to about 0.5 mm, about 0.05 mm to about 1 mm, about 0.05 mm to about 10 mm, about 0.05 mm to about 50 mm, about 0.05 mm to about 100 mm, about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, or about 50 mm to about 100 mm. The term about, when used in reference to the height of the multidirectional channel, means 0.001 mm, 0.005 mm, 0.01 mm, 0.05 mm, 0.1 mm, 0.5 mm, 1 mm, 10 mm, or 50 mm.

[0101] The channel length may be the total distance traveled by the fluid while exposed to the coating. In some embodiments of the device, the multidirectional channel has a length of about 0.1 mm to about 10,000 mm. In some embodiments of the device, the multidirectional channel has a length of at least about 0.1 mm. In some embodiments of the device, the multidirectional channel has a length of at most about 10,000 mm. In some embodiments of the device, the multidirectional channel has a length of about 0.1 mm to about 0.5 mm, about 0.1 mm to about 1 mm, about 0.1 mm to about 10 mm, about 0.1 mm to about 50 mm, about 0.1 mm to about 100 mm, about 0.1 mm to

about 500 mm, about 0.1 mm to about 1,000 mm, about 0.1 mm to about 5,000 mm, about 0.1 mm to about 10,000 mm, about 0.5 mm to about 1 mm, about 0.5 mm to about 10 mm, about 0.5 mm to about 50 mm, about 0.5 mm to about 100 mm, about 0.5 mm to about 500 mm, about 0.5 mm to about 1,000 mm, about 0.5 mm to about 5,000 mm, about 0.5 mm to about 10,000 mm, about 1 mm to about 10 mm, about 1 mm to about 50 mm, about 1 mm to about 100 mm, about 1 mm to about 500 mm, about 1 mm to about 1,000 mm, about 1 mm to about 5,000 mm, about 1 mm to about 10,000 mm, about 10 mm to about 50 mm, about 10 mm to about 100 mm, about 10 mm to about 500 mm, about 10 mm to about 1,000 mm, about 10 mm to about 5,000 mm, about 10 mm to about 10,000 mm, about 50 mm to about 100 mm, about 50 mm to about 500 mm, about 50 mm to about 1,000 mm, about 50 mm to about 5,000 mm, about 50 mm to about 10,000 mm, about 100 mm to about 500 mm, about 100 mm to about 1,000 mm, about 100 mm to about 5,000 mm, about 100 mm to about 10,000 mm, about 500 mm to about 1,000 mm, about 500 mm to about 5,000 mm, about 500 mm to about 10,000 mm, about 1,000 mm to about 5,000 mm, about 1,000 mm to about 10,000 mm, or about 5,000 mm to about 10,000 mm. The term about, when used in reference to length of the multidirectional channel, means 0.1 mm, 0.5 mm, 1 mm, 10 mm, 50 mm, 100 mm, 500 mm, 1,000 mm, or 5,000 mm. In some embodiments of the device, the multidirectional channel has a length of about 1.0 to about 3,000.0 mm

[0102] In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of about 1 mL/min to about 1,000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of at least about 1 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of at most about 1,000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates at a flow rate of about 1 mL/min to about 5 mL/min, about 1 mL/min to about 10 mL/min, about 1 mL/min to about 50 mL/min, about 1 mL/min to about 100 mL/min, about 1 mL/min to about 500 mL/min, about 1 mL/min to about 1,000 mL/min, about 5 mL/min to about 10 mL/min, about 5 mL/min to about 50 mL/min, about 5 mL/min to about 100 mL/min, about 5 mL/min to about 500 mL/min, about 5 mL/min to about 1,000 mL/min, about 10 mL/min to about 50 mL/min, about 10 mL/min to about 100 mL/min, about 10 mL/min to about 500 mL/min, about 10 mL/min to about 1,000 mL/min, about 50 mL/min to about 100 mL/min, about 50 mL/min to about 500 mL/min, about 50 mL/min to about 1,000 mL/min, about 100 mL/min to about 500 mL/min, about 100 mL/min to about 1,000 mL/min, or about 500 mL/min to about 1,000 mL/min. The term about, when used in reference to the flow rate, means 1 mL/min, 5 mL/min, 10 mL/min, 50 mL/min, 100 mL/min, 200 mL/min, 500 mL/min, or 1000 mL/min. In some embodiments of the device, the helical shaped multidirectional channel operates with flow rates of about 50 to about 400 mL/min.

[0103] In some embodiments, the detoxifying capacity of the devices, systems, and methods herein, alternatively called capture efficiency herein, as defined as the percent reduction of pathogenic bacteria at 1 minute compared to the starting concentration of the pathogenic bacteria in the fluid passed through the device is at least 30%, at least 35%, at least 40%, at least 45%, at least 50%, at least 55%, at least 60%, at least 65%, at least 70%, at least 75%, at least 80%, at least 81%, at least 82%, at least 83%, at least 84%, at least 85%, at least 86%, at least 87%, at least 88%, at least 89%, at least 90%, at least 91%, at least 92%, at least 93%, at least 94%, at least 95%, at least 96%, at least 97%, at least 98%, or at least 99%, at least about 30%, at least about 35%, at least about 40%, at least about 45%, at least about 50%, at least about 55%, at least about 60%, at least about 65%, at least about 70%, at least about 75%, at least about 80%, at least about 81%, at least about 82%, at least about 83%, at least about 84%, at least about 85%, at least about 86%, at least about 87%, at least about 88%, at least about 89%, at least about 90%, at least about 91%, at least about 92%, at least about 93%, at least about 94%, at least about 95%, at least about 96%, at least about 97%, at least about 98%, or at least about 99%, depending on the embodiment.

[0104] FIG. 8A and FIG. 8B illustrate a non-limiting embodiment of a multidirectional fluidic

channel with a spiral shaped channel. The multidirectional channel **801** may be spiral shaped. In some embodiments, spiral-shaped multidirectional channel **800** has two, 3 to 10-loop spiral channels **801** joined at an S-junction **802** to form a double spiral channel with one inlet **803** and one outlet **804** for bacterial/endotoxin removal. In some embodiments, the spiral-shaped multidirectional channel has two, 3 to 6-loop spiral channels joined at an S-junction to form a double spiral channel with one inlet and one outlet for bacterial/endotoxin removal (FIG. **8A** and FIG. **8B**). In some embodiments, the helical-shaped multidirectional channel comprises less than 3 loops with one inlet and one outlet for bacterial/endotoxin removal. In some embodiments, the helical-shaped multidirectional channel comprises more than 20 loops with one inlet and one outlet for bacterial/endotoxin removal. Ideally, the helical-shaped multidirectional channel is configured to provide maximum exposure in both time and surface area to enable it to capture, adsorb, and/or remove disease-causing material from biological fluids. Alternatively, the ultimate length and/or number of loops must be monitored to assure that an excessive pressure drop does not occur when the device is used in an in-line procedure with a patient.

[0105] The multidirectional channel may be a part of a device for the capture and adsorption of blood-borne materials as described herein. The multidirectional channel may comprise at least one inlet **803** and at least one outlet **804**. The multidirectional channel may be a double-spiral channel **800** joined at an S-junction **802**. FIG. **8A** illustrates a non-limiting embodiment of a spiral-shaped multidirectional channel, (channel Length=1,816 mm, Width=5.08 mm, Height=1.75 mm), fabricated using milling techniques and sealed by bolting two polycarbonate pieces together. FIG. **8B** illustrates another non-limiting embodiment of a spiral-shaped multidirectional channel, (channel Length=2,009 mm, Width=2 mm, Height=0.25 mm), fabricated using milling techniques and sealed by bolting two polycarbonate pieces together.

[0106] FIG. **9A** illustrates a non-limiting example of a helical-shaped multidirectional channel fabricated around a cylindrical chamber or mandrel. FIG. **9B** illustrates a helical-shaped multidirectional channel not fabricated around a cylindrical chamber or mandrel. FIG. **9C** illustrates a more compact helical-shaped multidirectional channel not fabricated around a cylindrical chamber or mandrel. The helical-shaped multidirectional channel device **900** is fabricated or extruded around or within a cylindrical chamber or mandrel (FIGS. **9A**, **9B**, **9C**, **22A**, **22B**, **23B**, **24A**, **24B**, **25A**, **25B** and **25C**). The vertically designed device offers a constant radius of curvature and compact size and shape. A single, assembled helical-shaped multidirectional channel **901** in some embodiments has 3 to 40 loops fabricated around a mandrel with one inlet **902** and one outlet **903** for bacterial/endotoxin removal. In some embodiments, the helical-shaped multidirectional channel is composed of 3 to 20-loops fabricated around a mandrel with one inlet and one outlet for bacterial/endotoxin removal. In some embodiments, a single assembled helical-shaped channel is connected in series to another single assembled helical-shaped channel using connector tubing within housing (FIG. **22B**). This allows for the capture and removal of Gram-negative, Gram-positive, and endotoxins from fluids within a single device.

[0107] FIG. **17** illustrates a non-limiting example of a helical-shaped multidirectional channel fabricated around a cylindrical chamber or mandrel that is sealed and enclosed using an outer sleeve. A helical-shaped multidirectional channel device **1700**, comprising a helical-shaped multidirectional channel **1702** fabricated around a cylindrical chamber or mandrel **1701**, by one of the methods described above, then is sealed and enclosed using an outer sleeve **1710**. FIG. **18** illustrates the channel(s)'s inlet(s) and/or outlet(s) fitting of the channel of FIG. **17**. Subsequent to enclosing the helical-shaped multidirectional channel in the sealed sleeve **1810**, the channel **1802** is fitted with appropriate inlet (i.e.: **1801**) and outlet (not shown) fittings as illustrated in FIG. **18**.

[0108] FIG. **21A** illustrates a non-limiting embodiment of a device comprising a filter plate that is a helical-shaped channel fabricated using machining methods or injection molding, with each filter plate channel being 98 mm length at the outermost edge of curve. In some embodiments, as illustrated in FIG. **21A**, a helical-shaped filter plate channel **2102** is fabricated using machining

methods or injection molding, as described above, with each filter plate channel being 98 mm in length at the outermost edge of its curve. Expanding this concept, multiple filter plates **2100**, **2102** are assemblable, stacked, and sealed together using solvent welding and adhesive, yielding a total channel length of 1,960 mm, as illustrated in FIG. **21B**, to create another variant of the claimed device, also comprising an inlet adapter **2101** and outlet **2103** adapter, each attached to the helical-shaped filter plate channel **2102**. FIG. **21C** illustrates a cross-sectional view of an embodiment comprising stacked helical plates **2100**, **2102**. FIG. **21D** illustrates the sealed joints **2109** on either side of each filter plate channel **2102** of an embodiment comprising stacked helical filter plates. Each plate may be made of translucent polycarbonate.

[0109] Expanding on the stacked plate configuration described herein, FIG. **22A** illustrates embodiment stacked helical filter plates in a sealed filter housing assembly **2200** and incorporating an inlet **2201** and outlet **2203** adapter each attached an inlet line and outlet line leading to or away from to the helical-shaped filter plate channel **2202**. The assembly is housed in a cylinder **2210** around a mandrel **2220** and the entire sub-assembly is capped **2207a**, **2207b** within the cylinder. FIG. **22A** illustrates a non-limiting embodiment of a stacked plate configuration described herein, (channel Length=1,960 mm, diameter of assembled device=4.44 cm, number of staked spirals=20, channel Length per plate=98 mm). In the embodiment illustrated in FIG. **22A**, the stacked plates may comprise 20 of the plates of FIG. **21A** being stacked on each other and glued together. In some embodiments, the helical shaped multidirectional stackable plate configuration is assembled using bolts, adhesive, binding material, thermal expansion, resin, epoxy, an inner sleeve, an outer sleeve, a base plate, a mandrel, cover glass, curing, extrusion welding, contact welding, high frequency welding, friction welding, laser welding, ultrasonic welding, solvent welding, or casting. In some embodiments, the stackable plate configuration is between 1 plate and 25 plates. In some embodiments, the stackable plate configuration is between 2 plate and 23 plates. In some embodiments, the stackable plate configuration is between 3 plate and 20 plates. FIG. **28** shows a line drawing of an assembled device of the embodiment illustrated in FIG. **22A**.

[0110] Further still, FIG. **22B** illustrates an embodiment comprising two sets of stacked helical filter plates **2202a**, **2202b**, in series, in a sealed housing **2210** spaced apart by a spacer tube **2220** and with the stacked helical filter plates **2202a**, **2202b** connected by a connector tube **2230**. The set of staked helical filter places **2202a** and **2202b** may be a variation or embodiment of the stacked plate configuration of FIG. **21B**. The first set of stacked helical filter plates **2202a** can be functionalized with one polypeptide such as polymyxin E, while the second set of helical plates **2202b** can be functionalized with another polypeptide such a vancomycin. This allows for the capture and removal of Gram-negative, Gram-positive, and endotoxins from flowing fluids within a single device. The embodiment of FIG. **22B** may be incorporated into a filtration system described herein.

[0111] FIG. **23A** illustrates an embodiment device comprising polycarbonate tubing extruded to form a helical-shaped multidirectional channel **2302**. Whereas, FIG. **23B** illustrates an embodiment device comprising polycarbonate tubing extruded to form a helical-shaped multidirectional channel **2302** that is fabricated around a mandrel **2320**. In some embodiments, the multidirectional channel is extruded polycarbonate tubing. In some embodiments, the multidirectional channel is enclosed. In some embodiments, the multidirectional stackable plate configuration is enclosed within a cylinder. In some embodiments, the multidirectional channel is extruded around a mandrel. The embodiment of FIG. **23A** and FIG. **23B** may comprise a multidirectional fluidic channel with similar dimensions, e.g. channel length, pitch, diameter, etc. of other multidirectional fluidic channels described herein.

[0112] FIG. **24A** illustrates an embodiment device made from a manufacturing method comprising forming an injection molded polycarbonate sheet **2405** attached to a mandrel **2420** with the helical-shaped channel **2402** incorporated into the sheet design. The formed sheet is then rolled into a helical shape around a mandrel by first heating the polycarbonate flat sheet and then tightly rolled

around the mandrel to create a helical shape. In some embodiments, two of these rolled sheets **2402a**, **2402b** can be connected in series using connector tubing (not shown), as illustrated in FIG. **24B**, placed in a cylinder **2410** with endcaps **2407a**, **2407b** having inlet **2401** and outlet **2403** adapters forming another assembly embodiment **2400**. In some embodiments, the sheet is rolled around the mandrel between 1 and 100 times. In some embodiments, the sheet is rolled around the mandrel between 1 and 20 times. In some embodiments, the sheet is rolled around the mandrel between 1 and 5 times. FIG. **24A** and FIG. **24B** may comprise a multidirectional fluidic channel with similar dimensions, e.g. channel length, channel width, etc. of other multidirectional fluidic channels described herein.

[0113] In yet another embodiment **2500**, injection molded polycarbonate parts **2502.sub.x1**, **2502.sub.x2** that incorporate the helical-shaped channel design into the housing can be generated with still another manufacturing method, then assembled, around a solid inner mandrel **2520** and sealed using ultrasonic welding and/or solvent welding **2511**, as illustrated in FIGS. **25A-25C**. FIG. **25A** illustrates an embodiment device made from injection molded polycarbonate parts that incorporate the helical-shaped channel design into the housing, while using a solid inner mandrel. FIG. **25B** illustrates a cross-sectional view of the helical housing wrapped around the mandrel of FIG. **25A**. FIG. **25C** illustrates an embodiment helical housing of FIG. **25A** assembled around the mandrel using ultrasonic welding and/or solvent welding. FIG. **25A**, FIG. **25B**, and FIG. **25C** may comprise a multidirectional fluidic channel with similar dimensions, e.g. channel length, pitch, diameter, etc. of other multidirectional fluidic channels described herein.

[0114] In some embodiments, the device comprises an outer tube that comprises a cast polycarbonate sleeve or extruded polycarbonate sleeve, and at least two cores, **2602a** (“A”) and **2602b** (“B”), comprising injection molded polycarbonate with the helical channel design incorporated into exterior of the cores. The faces of the at least two cores A and B, in some embodiments, is coated with elastomer, as illustrated in FIG. **26A-FIG. 26B**. Follow the coating step, the at least two cores **2602a** and **2602b**, (halves, as shown) are formed into an assembly **2640** and pressed into the outer tube or sleeve **2710** and sealed using an elastomer. Endcaps with inlet and outlet connections are then assembled to the ends of the tube or sleeve. FIG. **26A** and FIG. **26B** may comprise a multidirectional fluidic channel with similar dimensions, e.g. channel length, pitch, diameter, etc. of other multidirectional fluidic channels described herein.

[0115] Extending the prior concepts described in FIGS. **22B** and **24B**, FIG. **27A** illustrates two stacked and assembled injection molded polycarbonate parts (e.g.: **2640a**, **2640b**) that incorporate the helical-shaped channel design into the core pieces (**2640a/2602a** and **2640b/2602b**), while using a solid outer tube **2710**. The first piece of injection molded polycarbonate containing a helical channel **2640a/2602a**, can be functionalized with one polypeptide such as polymyxin E, while the second piece of injection molded polycarbonate containing a helical channel **2640b/2602b** can be functionalized with another polypeptide such a vancomycin. These two pieces of injection molded polycarbonate helical channels can be connected using a connector tube. This allows for the capture and removal of Gram-negative, Gram-positive, and endotoxins from flowing fluids within a single device. FIG. **27B** illustrates the cross-sectional view of the assembled helical-core channel device of FIG. **27A** with endcaps **2707a**, **2707b** having inlet **2701** and outlet **2703** adapters forming another assembly embodiment **2700**.

[0116] The claimed device can be made through the employment of any number of technologies. For example, it may be made through 3-D printing, soft lithography, photolithography, injection molding, blow molding, casting, ultrasonic welding, high frequency welding, heated tool or plate welding, solvent bonding, laser welding, spin welding, infrared welding, vibration welding, adhesive bonding, extrusion, and machining. In some embodiments, the machining methods are turning, drilling, boring, reaming, electric discharge machining and/or milling. Material may be machined or injection molded to create an enclosure of desired dimensions, such as by injection molding, extrusion, machining channel(s) as a whole or by machining halves that may be sealed

and enclosed using bolts, adhesive, epoxy, welding, thermal expansion, or another means (FIGS. 8A, 8B, 9A, 9B, 9C, 17, 18, 21A, 21B, 21C, 21D, 22A, 22B, 23A, 23B, 24A, 24B, 25A, 25B, 25C, 26A, 26B, 27A, 27B, and 28). For the channel(s), tubing may be attached to the inlet(s) and outlet(s) by the use of fittings, caps, or barbed luer lock connectors (FIGS. 18, 22A, 22B, 24A, 24B, 27A, 27B, and 28). The machining may be performed on various biocompatible materials, including various grades of polycarbonate. The device may then be coated with a substance, such as polymyxin or vancomycin, using previously described methods. The coating can coat at least a portion of an inner wall of the multidirectional channel(s) through conventional means.

[0117] In some embodiments of the device, the multidirectional channel is enclosed. In some embodiments of the device, the multidirectional channel is enclosed using bolts, adhesive, binding material, thermal expansion, resin, epoxy, an inner and/or outer sleeve, a base plate, a mandrel, cover glass, curing, extrusion welding, contact welding, high frequency welding, friction welding, laser welding, ultrasonic welding, solvent welding, or casting. In some embodiments of the device, the multidirectional channel is fabricated using at least one method selected from the group consisting of: 3-D printing, soft lithography, photolithography, injection molding, blow molding, casting, ultrasonic welding, high frequency welding, heated tool or plate welding, solvent bonding, laser welding, spin welding, infrared welding, vibration welding, adhesive bonding, machining, turning, drilling, boring, reaming, electric discharge machining, or milling. In some embodiments of the device, tubing is attached to the multidirectional channel inlet(s) and/or outlet(s) by the use of fittings, caps, or luer lock connectors.

[0118] Provided herein is a system for the capture and adsorption of blood-borne materials of interest comprising a fluidic cartridge with at least one inlet and at least one outlet; a multidirectional fluidic channel between the at least one inlet and the at least one outlet; said multidirectional fluidic channel comprising at least one inner wall; and a substance selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents, coating, fixed and covalently bonded to at least a portion of the at least one inner wall of the multidirectional fluidic channel. In some embodiments of the system, the substance coating the channel wall comprises a fixed, covalently-bonded polypeptide antibiotic. In some embodiments of the system, the covalently-bonded polypeptide antibiotic is polymyxin. In some embodiments, the covalently-bonded polypeptide antibiotic is vancomycin. In some embodiments of the system, the substance coating the channel wall comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine. In some embodiments of the system, the multidirectional channel is composed of at least one thermoplastic polymer base material that has at least one surface exposed functional group. In some embodiments of the system, the thermoplastic polymer base material has at least one surface exposed functional group selected from the groups consisting of: carbonyl groups, carboxyl groups, alcohol groups, amino groups, chloride groups, styrene groups, alpha-halogenated acyl group, benzyl groups, isocyanic acid groups, and other polymers or copolymers such as vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride and methylmethacrylate. In some embodiments of the system, the base material is polycarbonate. In some embodiments of the system, the fluidic device is disposable. The system may comprise any embodiment of the multidirectional fluidic channel described herein. The system may comprise the multidirectional fluidic channel of one or more of FIGS. 8A, 8B, 9A, 9B, 9C, 17, 18, 21A, 21B, 21C, 21D, 22A, 22B, 23A, 23B, 24A, 24B, 25A, 25B, 25C, 26A, 26B, 27A, 27B, and 28.

[0119] In some embodiments, the claimed device is configurable for in-line filtration and capture such that whole blood infected with bacteria and/or endotoxin **1906** can be pumped **1930** from a patient to the inlet **1901** of the double-spiral fluidic device **1900**, as illustrated in FIG. 19. The

device is functionalized along the channel walls **1902** with polypeptide antibiotics designed to capture bacteria and endotoxin. Following processing through the fluidic device, the blood then containing only healthy material is returned to the patient via the outlet **1903**.

[0120] Similarly, as illustrated in FIG. **20**, is an embodiment configurable for in-line filtration and capture such that whole blood infected with bacteria and/or endotoxin can be pumped **2010** from a patient to the inlet **2001** of the helical fluidic devices in series **2000**. The embodiment devices are functionalized along the channel walls **2002** with polypeptide antibiotics designed to capture bacteria and endotoxin **2006**. Following processing through the fluidic devices, the blood then containing only healthy material is returned to the patient via the outlet **2003**.

[0121] In some embodiments, bacteria and endotoxin are removed from blood by circulation outside the body wherein prior to return to the patient body, blood removed from the patient body or plasma components separated from blood are passed through the multidirectional bacterial and endotoxin adsorbent device as described herein, and the cleaned blood or plasma components from which bacteria and endotoxin were removed are returned to the patient body (FIG. **19**).

[0122] Moreover, it should be appreciated that any of the devices described or contemplated herein may be used in series or parallel (FIG. **20**).

[0123] Provided herein is a method for the capture and adsorption of materials of interest comprising bringing a sample of fluid in contact with a multidirectional, polypeptide antibiotic coated channel; adsorbing said material(s) of interest on said multidirectional channel walls; and detecting the presence or amount of said materials of interest captured within the multidirectional, polypeptide antibiotic coated channel. In some embodiments of the method, a fluid sample is brought in contact with a multidirectional, polypeptide antibiotic coated channel using a pump. The method may comprise any variant, embodiment, or example of the channels described herein. In some embodiments, the pump is a peristaltic pump. In some embodiments, the pump is a syringe pump. In some embodiments of the method, a fluid sample is brought in contact with a multidirectional, polypeptide antibiotic coated channel using a syringe. In some embodiments, the polypeptide antibiotic is polymyxin. In some embodiments, the polypeptide antibiotic is vancomycin. In some embodiments of the method, heparin, sodium citrate, or other anticoagulants, are added to the fluid sample prior to being brought in contact with a multidirectional, polypeptide antibiotic coated channel. In some embodiments, detection of the presence or amount of materials of interest captured within the multidirectional, polypeptide antibiotic coated channel is performed using a method selected from the group comprising: cell counting, MALDI-TOF MS (matrix assisted laser desorption ionization-time of flight mass spectrometry), mass spectrometry, PCR (polymerase chain reaction), biosensing, flow cytometry, and fluorescent labeling.

[0124] Provided herein is a method for treating a patient suspected of having bacteremia, endotoxemia or sepsis comprising contacting a sample of blood with the fluidic cartridge containing a multidirectional fluidic channel between at least one inlet and at least one outlet; adsorbing one or more materials of interest on at least one wall of the multidirectional fluidic channel; removing the one or more materials of interest from the sample of blood to produce treated blood; and returning said treated blood to said patient. The method may comprise any variant, embodiment, or example of the channels described herein.

[0125] Provided herein is a method for diagnosing a patient suspected of having bacteremia, endotoxemia or sepsis comprising contacting a sample of fluid with a fluidic cartridge containing a multidirectional fluidic channel between at least one inlet and at least one outlet; adsorbing one or more materials of interest on at least one wall of the multidirectional fluidic channel; and detecting the presence or amount of materials of interest captured along the at least one wall of the multidirectional fluidic channel. The method may comprise any variant, embodiment, or example of the channels described herein. In some embodiments, the method comprises identifying the captured materials of interest through a procedure selected from the group consisting of: polymerase chain reaction (PCR), fluorescence in situ hybridization (FISH), optically active

microbeads, optically active nanoparticles, and matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF). In some embodiments, the method comprises identifying the captured materials of interest through culturing methods. In some embodiments, the method comprises identifying the captured materials of interest through elution of the captured materials of interest followed by a procedure selected from the group consisting of polymerase chain reaction (PCR), fluorescence in situ hybridization (FISH), optically active microbeads, optically active nanoparticles, and/or matrix-assisted laser desorption/ionization time-of-flight (MALDI-TOF), or culturing.

[0126] Provided herein are materials used for removal of pathogens and endotoxins from fluids for transfusion medicine, sepsis treatment, bacteremia treatment, endotoxemia treatment, and other blood-borne diseases. These fluids, such as blood, may be passed or circulated through the channel(s) of the bacterial and endotoxin adsorbent device. The method may comprise any variant, embodiment, or example of the channels described herein.

[0127] Provided herein are methods in which thermoplastic tubing is coated with the polypeptide antibiotic substance. The method may comprise any variant, embodiment, or example of the channels described herein.

EXAMPLES

[0128] It is understood that the examples and embodiments described herein are for illustrative purposes only and are not intended to limit the scope of the claimed invention. It is also understood that various modifications or changes in light of the examples and embodiments described herein will be suggested to persons skilled in the art and are to be included within the spirit and purview of this application and scope of the appended claims.

[0129] The following examples utilize spiral-shaped multidirectional channel devices with the following parameters (channel Length=1,816-2,009 mm, Width=2 mm to 5.08 mm, Height=0.25-1.75 mm), fabricated using milling techniques and sealed by bolting two polycarbonate pieces together. Hexamethylene diamine solution (10% (v/v) aqueous in 100 mM borate buffer) was added to the polycarbonate channel surface for 45 minutes. Addition of the diamine results in a surface decorated with amine groups that are available for subsequent functionalization reactions.

[0130] For colistinated devices, colistin sulfate salt (0.28 g) was added to 16 mLs of a fifty-fifty distilled water/ethanol mixture (50% DiH₂O/50% EtOH, v/v). The colistin solution was added to the polycarbonate-based channel and incubated within the channel at room temperature for 1 hour. Afterwards, the channel was washed with EtOH and air dried.

[0131] For devices with vancomycin, vancomycin (0.22 g) was added to 13 mLs of an ethanol/water mixture (50% DiH₂O/50% EtOH (v/v)). The vancomycin solution was added to the polycarbonate-based channel and incubated within the channel at room temperature for 1 hour. Afterwards, the channel was washed with EtOH and air dried.

[0132] For pegylated devices, NHS-PEG.sub.1000-NHS (0.24 grams (g)) was added to 16 mLs of an ethanol/water mixture (50% DiH.sub.2O/50% EtOH (v/v)). The NHS-PEG.sub.1000-NHS solution was added to the polycarbonate-based channel and incubated within the channel at room temperature for 45 minutes. Afterwards, the channel was washed with EtOH and dried.

[0133] FIG. 2A and FIG. 2B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel using *Acinetobacter baumannii*. *Acinetobacter baumannii* ATCC 17978 bacterial cells were chosen as a standard specimen for qualitatively and quantitatively evaluating the bacterial capture capacity of the colistinated and polyethylene glycol (PEGylated) functionalized fluidic devices. FIG. 2A illustrates adsorption of fluorescently labeled *A. baumannii* ATCC 17978 along polymyxin functionalized walls of an embodiment polycarbonate device. Fluorescently labeled *A. baumannii* ATCC 17978 are bound to the colistinated channel walls following passage through the fluidic device at 80 mL min.sup.-1 and subsequent washing (FIG. 2A). FIG. 2B illustrates absence of adsorption of fluorescently labeled *A. baumannii* ATCC 17978 along PEGylated walls of an embodiment

polycarbonate device. Significant bacterial capture did not occur within the PEGylated spiral fluidic device, which served as a control, presumably due to the absence of colistin surface functionalization (FIG. 2B).

[0134] FIG. 3A and FIG. 3B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel for *Staphylococcus aureus*. *Staphylococcus aureus* ATCC 29213 bacterial cells were chosen as a standard specimen for qualitatively and quantitatively evaluating the bacterial capture capacity of the vancomycin functionalized and Polyethylene glycol (PEGylated) functionalized fluidic devices. FIG. 3A illustrates adsorption of fluorescently labeled *S. aureus* ATCC 29213 along vancomycin functionalized walls of an embodiment polycarbonate device. Fluorescently labeled *S. aureus* ATCC 29213 bacterial cells are bound to the vancomycin functionalized channel walls following passage through the fluidic device at 80 mL min.sup.-1 and subsequent washing (FIG. 3A). FIG. 3B illustrates absence of adsorption of fluorescently labeled *S. aureus* ATCC 29213 along PEGylated walls of an embodiment polycarbonate device. Significant bacterial capture did not occur within the PEGylated spiral fluidic device, which served as a control, presumably due to the absence of vancomycin surface functionalization (FIG. 3B).

[0135] FIG. 10A and FIG. 10B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel using *Acinetobacter baumannii*. FIG. 10A illustrates a capture of *A. baumannii* ATCC 17978 when flowed through the polymyxin coated device (i.e. colistinated device) at 80 ml min.sup.-1. The *A. baumannii* ATCC 17978 concentration flowed through the device was approximately 105 colony-forming units per milliliter (CFU/mL). No capture was observed within the PEGylated device. n=4, p<0.05*. Bacterial capture capacity, thus, was quantified by flowing *Acinetobacter baumannii* ATCC 17978 suspensions through the colistin coated fluidic device at a flow rate of 80 mL min.sup.-1. A bacterial cell capture capacity of over 107 colony forming units (CFUs) was achieved with a single passage through the colistinated double spiral fluidic device and a capture efficiency of nearly 100% within the first minute of flow (FIG. 10A and FIG. 10B), as confirmed by the reduction of colonies that could be cultured from the fluid exiting the device. FIG. 10B illustrates the total capture capacity of the device was 2.39E7 CFUs.

[0136] FIG. 11A and FIG. 11B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel for *Klebsiella pneumoniae*. FIG. 11A illustrates a capture of *K. pneumoniae* ATCC 700603 when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min.sup.-1. The *K. pneumoniae* ATCC 700603 concentration flowed through the device was approximately 105 CFU/mL. n=4, p<0.05*. *Klebsiella pneumoniae* ATCC 700603, another Gram-negative human pathogen, was also successfully removed from flowing fluid with an associated capture efficiency of nearly 100% within the first minute of flow through the colistinated device (FIG. 11A and FIG. 11B). FIG. 11B illustrates the total capture capacity of the device was 1.98E7 CFUs.

[0137] FIG. 12A and FIG. 12B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel using *Acinetobacter baumannii*. The colistinated double spiral fluidic device also captured and removed antibiotic-resistant organisms, including colistin-resistant *A. baumannii* (Qureshi, Z. A. et al. Colistin-resistant *Acinetobacter baumannii*: Beyond carbapenem resistance. Clin. Infect. Dis. 60, 1295-1303 (2015)) and colistin-resistant *K. pneumoniae* ATCC 700603, from flowing fluid (FIGS. 12A, 12B, 13A, and 13B). FIG. 12A illustrates the capture of colistin resistant *A. baumannii* when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min.sup.-1. The colistin resistant *A. baumannii* concentration flowed through the device was approximately 10.sup.4 CFU/mL, n=4, p<0.05*. FIG. 12B illustrates the initial capture efficiency was nearly 100%; the total capture capacity of the device was 7.89E6 CFUs.

[0138] FIG. 13A and FIG. 13B show example implementations of a device of the present

disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel for *Klebsiella pneumoniae*. FIG. 13A illustrates the capture of colistin resistant *K. pneumoniae* ATCC 700603 when flowed through the polymyxin E coated device (i.e. colistinated device) at 80 ml min.^{sup.}-1. The colistin resistant *K. pneumoniae* ATCC 700603 concentration flowed through the device was approximately 10.^{sup.}4 CFU/mL, n=4, p<0.05*. FIG. 13B illustrates the initial capture efficiency was nearly 80%. The total capture capacity of the device was 7.31E6 CFUs.

[0139] FIG. 14A and FIG. 14B show example implementations of a device of the present disclosure to evaluate the bacterial capture capacity a multidirectional fluidic channel for *Staphylococcus aureus*. The vancomycin functionalized double spiral fluidic device captured and removed Gram-positive pathogens from flowing fluid, including *Staphylococcus aureus* ATCC 29213, with a capture efficiency of nearly 80% within the first minute of flow (FIGS. 14A and 14B). FIG. 14A illustrates the capture of *S. aureus* ATCC 29213 when flowed through the vancomycin functionalized device at 80 ml min.^{sup.}-1. The *S. aureus* ATCC 29213 concentration flowed through the device was approximately 10.^{sup.}4 CFU/mL. No capture was observed within the PEGylated device. n=4, p<0.05*. FIG. 14B illustrates the total capture capacity of the device was 5.24E6 CFUs. PEGylated devices did not capture Gram-positive *S. aureus* ATCC 29213 or Gram-negative *A. baumannii* ATCC 17978, confirming specificity for pathogens in colistin- and vancomycin-functionalized devices (FIGS. 11B and 14B).

[0140] A summary of test data is presented in FIG. 15, which shows the capture capacity of the bacteria adsorbent device using a variety of bacterial strains as described and presented in FIGS. 10A-14B.

[0141] Endotoxin, one of the principal components of the outer membrane of Gram-negative bacteria, also contributes to the systemic inflammatory response that is characteristic of sepsis (Wiesel, P. et al. Endotoxin-induced mortality is related to increased oxidative stress and end-organ dysfunction, not refractory hypotension, in heme oxygenase-1-deficient mice. Circulation 102, 3015-3022 (2000)). Therefore, the removal of endotoxin from flowing fluid using embodiments of the multidirectional fluidic device described herein, according to methods herein was assessed. FIG. 16 illustrates that an endotoxin was spiked into endotoxin-free water (1 ug ml.^{sup.}-1) and flowed through the polymyxin E coated device (i.e. colistinated device) and PEGylated double spiral fluidic devices at 80 ml min.^{sup.}-1. The amount of endotoxin captured over time was assessed. Results were plotted as the mean±SD, n=3. Endotoxin was rapidly captured from the fluid passing through the colistinated fluidic device, with endotoxin capture efficiency approaching 100% in single pass operation, as shown in FIG. 16. In contrast, endotoxin was not effectively captured using the PEGylated device, demonstrating the critical requirement for colistin in endotoxin adsorption in the multidirectional fluidic device.

[0142] FIG. 28 shows a line drawing of an assembled device of the embodiment illustrated in FIG. 22A. Length of Channel –2060 mm (at outer wall): Pitch –3.8 mm: Radius of Curvature-16.5 mm (radius of outer wall): Diameter of Device –44.5 mm (OD of housing): Number of plates –20.

Claims

1.-4. (canceled)

5. A plate for the capture and adsorption of blood-borne materials of interest, the plate comprising: an inlet portion; an outlet portion; and a portion of a multidirectional channel fluidically connecting the inlet portion and the outlet portion, wherein the outlet portion is configured to interface with an inlet portion of a second plate such that the multifluidic channel is extended, wherein the portion of the multidirectional fluidic channel comprises a substance coating at least a portion of at least one inner wall of the portion of the multidirectional fluidic channel, and wherein the substance is effective to capture or to adsorb a blood-borne material.

6. The plate of claim 5, wherein the substance is effective to capture or to adsorb a Gram-negative

bacterium, a Gram-positive bacterium, or an endotoxin.

7. The plate of claim 5, wherein the portion of the multidirectional fluidic channel comprises a thermoplastic polymer base material that has at least one surface exposed functional group selected from the group consisting of: polycarbonate, vinylchloride, vinylacetate, acrylamide, polyethylene, polyethylene terephthalate acrylic acid, acrylonitrile, maleic anhydride, and methylmethacrylate.

8. The plate of claim 5, wherein the substance is selected from the group consisting of: antibodies, crosslinking agents, peptides, proteins, antibiotics, polymers, amines, polyethers, amino acids, aptamers, tumor necrosis factors, adhesion receptors, E-selectin, cytokines, chemotherapy agents, quorum sensing proteins, quorum sensing receptors, and biological agents.

9. The plate of claim 8, wherein the substance comprises a fixed, covalently-bonded antibiotic.

10. The plate of claim 9, wherein the fixed, covalently-bonded antibiotic is polymyxin or vancomycin.

11. The plate of claim 10, wherein an amount of the polymyxin or the vancomycin is at least 0.5 mM.

12. The plate of claim 5, wherein the substance is coating, fixed, and covalently bonded to at least a portion of the at least one inner wall of the portion of the multidirectional fluidic channel.

13. The plate of claim 5, wherein the substance comprises a fixed crosslinking agent selected from the group consisting of: hexamethylene diamine, polyethylene glycol, polyethylene glycol derivatives, N-hydroxysuccinimide esters, and glycine.

14. A method for capture and adsorption of materials of interest, the method comprising: flowing a sample of fluid through a multidirectional fluidic channel from at least one inlet to at least one outlet, wherein the multidirectional fluidic channel fluidically connects the at least one inlet and the at least one outlet, wherein the multidirectional fluidic channel comprises one or more plates arranged in a stackable plate configuration, and wherein the multidirectional fluidic channel comprises a substance coating at least a portion of at least one inner wall of the multidirectional fluidic channel; and adsorbing or capturing one or more blood-borne materials of interest the substance coating at least the portion of the at least one inner wall of the multidirectional fluidic channel.

15. The method of claim 14, further comprising: detecting a presence or amount of the one or more blood-borne materials of interest.

16. The method of claim 15, further comprising diagnosing a patient suspected of having bacteremia, endotoxemia, or sepsis based on the presence or amount of the one or more blood-borne materials of interest.

17. The method of claim 14, wherein the sample is a sample of blood.

18. The method of claim 17, further comprising: removing the one or more blood-borne materials of interest from the sample of blood to produce treated blood; and returning the treated blood to the patient.

19. The method of claim 17, wherein the sample is a sample of whole blood.

20. The method of claim 14, wherein the substance is effective to capture or to adsorb a Gram-negative bacterium, a Gram-positive bacterium, or an endotoxin.

21. The method of claim 14, further comprising: flowing the sample of fluid through a second multidirectional fluidic channel comprising at least one inner wall, wherein the substance coating the at least the portion of the at least one inner wall of the multidirectional fluidic channel comprises a first antibiotic, and wherein a second substance coating at least a portion of the at least one inner wall of the second multidirectional fluidic channel comprises a second antibiotic.

22. The method of claim 21, wherein the substance and the second substance are different substances.

23. The method of claim 14, wherein the stackable plate configuration comprises a number of plates within a range from 1 plate to 25 plates.

24. The method of claim 14, wherein the substance is effective to capture or to adsorb the blood-borne material with a capture capacity of 10^3 colony forming units or greater.
